

# Formalizing Bombieri Vinogradov

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# Chapter 1

## Introduction

Let  $\Lambda$  be the von Mangoldt function and, for a modulus  $q \geq 1$  and a residue class  $a \bmod q$ , let

$$\psi(x; q, a) := \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n)$$

be Chebyshev’s function in the arithmetic progression  $a \bmod q$ . The prime number theorem in arithmetic progressions predicts  $\psi(x; q, a) \approx x/\varphi(q)$  when  $(a, q) = 1$ . The best unconditional estimate for an *individual* modulus, the Siegel–Walfisz theorem [7, 11], only covers  $q \leq (\log x)^C$ , and even the generalised Riemann hypothesis would only reach  $q \leq x^{1/2-\varepsilon}$ . The theorem of Bombieri [1] and A. I. Vinogradov [10] shows that the latter range is available *on average* over the modulus: for every  $A \geq 0$ ,

$$\sum_{q \leq Q} \max_{y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A} \quad \text{uniformly for } x \geq 2 \text{ and } 1 \leq Q \leq \frac{\sqrt{x}}{(\log x)^{A+3}}.$$

This is often described as “the generalised Riemann hypothesis on average”. It is the main analytic input in sieve-theoretic applications such as the work of Goldston–Pintz–Yıldırım and Maynard on small gaps between primes, and it is the reason the level of distribution  $1/2$  plays such a prominent role in that subject.

The original proofs used the large sieve together with zero-density estimates for Dirichlet  $L$ -functions. The proof formalised here is the elementary one due to Vaughan [8, 9], in the form presented in Chapter 26 of Koukoulopoulos [4] (see also [2, Ch. 28] and [3, §17.4]). Vaughan’s identity splits  $\Lambda$  into three pieces; one is small, one is a *Type I* sum that is bounded pointwise by elementary means, and one is a *Type II* (bilinear) sum that is bounded on average over  $q$  by the large sieve, after the small conductors have been handled with the Siegel–Walfisz theorem.

## What is formalised

This blueprint describes a formalisation of the Bombieri–Vinogradov theorem in Lean 4 and Mathlib. Two analytic results are taken as external inputs, i.e. as axioms in Lean: the Siegel–Walfisz theorem (Theorem 1) and the multiplicative large sieve inequality (Theorem 2). Everything else is proved from Mathlib, apart from the smoothed cutoff function and its basic properties, which are borrowed from the PrimeNumberTheoremAnd library [6] (see Proposition 9.11).

*Theorem* (Main result, informal form). Assume the Siegel–Walfisz theorem and the large sieve inequality. Then for every  $A \in \mathbb{N}$ , all real  $x \geq 2$  and  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,

$$\sum_{q \leq Q} \sup_{1 \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A}.$$

The precise statement is Theorem 3.

## How to read this document

The text is organised as a paper, not as a commentary on the Lean code. Each numbered statement carries exactly the hypotheses that the formal statement carries, and each proof is complete modulo earlier numbered results; the proofs do, however, follow the route taken by the formalisation, which occasionally differs from the textbook route (Section 2.4). Below each statement, the line *Lean* lists the declarations that formalise it. A single mathematical statement is frequently spread over several Lean declarations, for instance a pointwise bound and its maximal or summed form, or an identity and the two or three special cases in which it is applied. A few standard facts that are quoted from Mathlib or from the PrimeNumberTheoremAnd library appear as numbered results with a one-line proof saying so, so that no proof in this document appeals to an unnamed “standard fact”.

Implied constants are not tracked: in Lean every  $\ll$  below is witnessed by some real number, obtained where convenient by a choice or a compactness argument, and nothing is claimed about its size. In particular the implied constant in the Siegel–Walfisz axiom is not effective, and neither is any constant that depends on it. Notation 3.1 fixes the conventions.

## Chapter 2

# Overview of the proof

### 2.1 Discrepancies and the reduction to $\Lambda$

For an arithmetic function  $f$  we measure its distribution in progressions by the *discrepancy*

$$\Delta_f(y; q, a) := \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} f(n) - \frac{1}{\varphi(q)} \sum_{\substack{n \leq y \\ (n, q) = 1}} f(n)$$

(Definition 3.17). For  $f = \Lambda$  the difference between  $\psi(y; q, a) - y/\varphi(q)$  and  $\Delta_\Lambda(y; q, a)$  is  $\varphi(q)^{-1}(\sum_{n \leq y, (n, q) = 1} \Lambda(n) - y)$ , which is controlled by the prime number theorem with logarithmic savings (a special case of Siegel–Walfisz, Corollary 4.1) together with the elementary bound  $\sum_{n \leq y, (n, q) > 1} \Lambda(n) \ll \log q \log y$  (Lemma 3.7). Summing  $1/\varphi(q)$  over  $q \leq Q$  costs a factor  $\log x$  (Lemma 5.3). The main theorem is thus reduced to a statement about  $\Delta_\Lambda$  (Proposition 10.1), in which the maximum over  $y$  is restricted to  $\sqrt{x} \leq y \leq x$ ; the range  $y < \sqrt{x}$  is handled by the trivial bound  $|\psi(y; q, a) - y/\varphi(q)| \ll y$  (Lemma 3.6).

### 2.2 Vaughan’s identity

Fix parameters  $U, V$  with  $UV \leq \sqrt{x}$  and  $U, V \geq e^{\sqrt{\log x}}$  (Definition 3.23). Vaughan’s identity (Proposition 6.2) decomposes

$$\Lambda = \Lambda^\sharp + \Lambda^\flat + \Lambda_{\leq U}, \quad \Lambda^\sharp = \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1, \quad \Lambda^\flat = (\Lambda_{> U} * 1) * \mu_{> V}.$$

The three pieces are treated separately, in Chapter 7, Chapter 8 and Chapter 9 respectively, and recombined by the triangle inequality in Chapter 10. The piece  $\Lambda_{\leq U}$  is supported on  $[1, U]$  and is bounded trivially. The piece  $\Lambda^\sharp$  is a *Type I* sum: a convolution of a coefficient function of small support with the smooth functions  $\log$  and  $1$ , whose discrepancies are bounded pointwise by Abel summation and  $|\Delta_1| \leq 1$ . The piece  $\Lambda^\flat$  is a *Type II* (bilinear) sum, which is bounded on average over  $q$  by expanding  $\Delta_{\Lambda^\flat}$  in Dirichlet characters, grouping characters by conductor, treating the small conductors with Siegel–Walfisz, and treating the large conductors with the large sieve.

## 2.3 The two external inputs

*Siegel–Walfisz* (Theorem 1) is used in three places: with  $q = 1$  as the prime number theorem (Corollary 4.1), in the reduction from  $\psi$  to  $\Delta_\Lambda$  (Lemma 10.4), and in the treatment of the small conductors  $d \leq (\log x)^C$  in the Type II sum (Lemma 9.5). The *large sieve* (Theorem 2) enters only through the maximal large sieve inequality for convolutions (Proposition 9.21, Theorem 26.6 of [4]), which we prove via a smoothed Perron formula rather than the truncated Perron formula of the textbook (Section 9.4).

## 2.4 Where the formalisation deviates from the textbook

The following points are worth keeping in mind when comparing with [4].

- *A divisor factor in the Type I bound.* The pointwise bound for the restricted function  $\Lambda_r^\sharp$  carries a factor  $\tau(r)$  (Proposition 8.7). Restriction to integers coprime to  $r$  commutes with convolution (Lemma 3.13), so the smooth factor  $\log$  becomes  $\log_r$ , which is neither smooth nor monotone, and we Möbius-expand it as a sum of  $\tau(r)$  dilated copies of  $\log$  and 1 (Lemma 8.6). The factor is harmless: in the Type I average it is summed over the modulus using  $\sum_{q \leq Q} \tau(q) \ll Q \log Q$  (Lemma 5.1), and in the small-conductor analysis it is absorbed by  $\tau(q) \leq 2\sqrt{q} \leq 2x^{1/4}$ , which is why the small-conductor lemmas assume  $q \leq \sqrt{x}$ .
- *The large sieve for convolutions is proved by smoothing.* Instead of the truncated Perron formula, we use the smoothed cutoff  $\tilde{1}_\varepsilon$  of the `PrimeNumberTheoremAnd` library (Definition 9.12). At half-integers  $y = K + \frac{1}{2}$  the smoothed and sharp cutoffs agree exactly when  $\varepsilon = 1/(6 \log 2 \cdot x)$  (Lemma 9.20), so no error term arises, and Mellin inversion on the line  $\Re s = 1/\log(x+1)$  (Lemma 9.16) reduces the problem to the large sieve for Dirichlet polynomials (Lemma 9.17) and to a kernel integral of size  $O(\log x)$  (Lemma 9.18).
- *Uniformity in  $U, V$ .* All Type I/II estimates are proved for an arbitrary parameter datum (Definition 3.23); the choice  $U = V = e^{\sqrt{\log x}}$  is only made in the final assembly. The constraints force  $\log x \geq 16$  (Lemma 3.24), so the main theorem splits into the range  $x \leq e^{32}$ , where the trivial bound suffices (Lemma 10.3), and the range  $x > e^{32}$ .
- *Maxima are suprema in  $[0, \infty]$ .* The maxima over  $y$  and  $a$  are formalised as suprema in the extended non-negative reals (Definition 3.25), so that no boundedness hypotheses need to be carried around; finiteness is part of each estimate. This is invisible in the statements below, where we simply write  $\max$ .

# Chapter 3

## Notation and preliminaries

This chapter fixes conventions, records the handful of standard facts from analytic number theory that the argument quotes from Mathlib, and introduces the two objects around which the whole proof is organised: the discrepancy  $\Delta_f(y; q, a)$  of an arithmetic function in a residue class, and the parameter datum  $(x, U, V)$  with respect to which all the estimates of Chapters 6 to 10 are uniform.

### 3.1 Conventions

**Notation 3.1** (Asymptotic notation and implied constants). For real-valued quantities  $f, g$  the statement  $f \ll g$  (equivalently  $f = O(g)$ ) means  $|f| \leq Cg$  for some constant  $C > 0$ ; a subscript, as in  $f \ll_A g$ , records the parameters the constant may depend on. Unless stated otherwise the constant may also depend on the implied constants of the two external inputs (Theorem 1, Theorem 2) and on the fixed bump function of Definition 9.12, but never on  $x, y, q, a, Q$  or on the parameter datum of Definition 3.23. Constants are not tracked in the formalisation: each  $\ll$  is witnessed by some real number and nothing is asserted about its size.

**Notation 3.2** (Arithmetic functions).  $\mathbb{N} = \{0, 1, 2, \dots\}$  denotes the natural numbers, although all sums over  $n \leq x$  start at  $n = 1$  (Definition 3.3). For  $a, b \in \mathbb{N}$  we write  $(a, b) = \gcd(a, b)$ , with the convention  $(0, b) = b$ . Dirichlet convolution is written  $f * g$ ;  $1$  denotes the constant function 1 on  $\mathbb{N}_{\geq 1}$  (so that  $f * 1$  is the divisor sum of  $f$ ),  $\delta_1$  its convolution identity,  $\mu$  the Möbius function,  $\varphi$  Euler's totient,  $\tau$  the divisor function and  $\Lambda$  the von Mangoldt function;  $\log$  also denotes the arithmetic function  $n \mapsto \log n$ , and  $\log^v$  its pointwise  $v$ -th power (with  $\log^0 = 1$ ). All arithmetic functions vanish at 0.

For  $q \in \mathbb{N}$  and  $a \in \mathbb{Z}/q\mathbb{Z}$  we write  $n \equiv a \pmod{q}$  for the natural numbers whose residue class modulo  $q$  is  $a$ ; when  $q = 0$  we identify  $\mathbb{Z}/0\mathbb{Z} = \mathbb{Z}$ , and when  $q = 1$  every natural number lies in the single residue class.  $(\mathbb{Z}/q\mathbb{Z})^*$  is the unit group, of order  $\varphi(q)$  for  $q \geq 1$ . For a Dirichlet character  $\chi$  modulo  $q$  we write  $\chi_0$  for the principal character,  $\sum_{\chi \bmod q}^*$  for a sum over primitive characters, and  $\xi \uparrow^q$  for the character modulo  $q$  induced by a character  $\xi$  modulo a divisor of  $q$ , i.e.  $(\xi \uparrow^q)(n) = \xi(n)1_{(n, q)=1}$ . Complex conjugation is  $\bar{z}$  and  $|z|$  is the absolute value, also for complex  $z$ .

**Definition 3.3** (Sums over  $n \leq x$ ). For  $f : \mathbb{N} \rightarrow R$  with values in an additive commutative monoid and a real number  $x$ , the *summatory function* of  $f$  is

$$\sum_{n \leq x} f(n) := \sum_{1 \leq n \leq \lfloor x \rfloor} f(n).$$

The term  $n = 0$  is never included, and the sum is empty (hence 0) when  $x < 1$ . Throughout, every sum of the form  $\sum_{n \leq x}$  with a real endpoint  $x$  is to be read in this way, and a sum  $\sum_{n \leq x, P(n)}$  with a side condition is the summatory function of  $f \cdot 1_P$ .

## 3.2 Chebyshev's function in arithmetic progressions

**Definition 3.4** (Chebyshev's function in progressions). For a real number  $x$ , a modulus  $q \in \mathbb{N}$  and a residue class  $a \in \mathbb{Z}/q\mathbb{Z}$ , set

$$\psi(x; q, a) := \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n).$$

We do not require  $a$  to be coprime to  $q$  in the definition. For  $q = 1$  this is the classical Chebyshev function  $\psi(x) = \sum_{n \leq x} \Lambda(n)$ .

**Lemma 3.5** (Chebyshev's bound). *For every real  $x \geq 0$ ,  $\psi(x) \ll x$ , with an absolute implied constant.*

*Proof.* This is Chebyshev's elementary estimate, available in Mathlib (with the explicit constant  $\log 4 + 4$ ).  $\square$

**Lemma 3.6** (Trivial bound for the Chebyshev error). *Let  $q \geq 1$ , let  $a \in \mathbb{Z}/q\mathbb{Z}$  and let  $z \geq 1$  be real. Then*

$$0 \leq \psi(z; q, a) \leq \psi(z) \quad \text{and} \quad \left| \psi(z; q, a) - \frac{z}{\varphi(q)} \right| \ll z,$$

*with an absolute implied constant.*

*Proof.* Every summand  $\Lambda(n)$  is non-negative, so  $\psi(z; q, a)$  is a sub-sum of  $\psi(z)$  and  $0 \leq \psi(z; q, a) \leq \psi(z)$ . Both  $\psi(z; q, a)$  and  $z/\varphi(q)$  are non-negative, so the absolute value of their difference is at most the larger of the two. By Lemma 3.5,  $\psi(z; q, a) \leq \psi(z) \ll z$ ; on the other hand  $\varphi(q) \geq 1$ , so  $0 \leq z/\varphi(q) \leq z$ .  $\square$

**Lemma 3.7** (Prime powers sharing a factor with  $q$ ). *For every real  $z \geq 2$  and every integer  $q \geq 1$ ,*

$$0 \leq \sum_{\substack{n \leq z \\ (n, q) > 1}} \Lambda(n) \leq \frac{\log q \cdot \log z}{\log 2}.$$

*Proof.* Non-negativity is clear. The summand vanishes unless  $n = p^k$  is a prime power, and  $\Lambda(p^k) = \log p$ . Grouping by the exponent  $k$ , which satisfies  $1 \leq k \leq \log z / \log 2$  (as  $2^k \leq p^k \leq z$ ), the sum equals

$$\sum_{1 \leq k \leq \log z / \log 2} \sum_{\substack{p \leq z^{1/k} \\ (p^k, q) > 1}} \log p.$$

For a prime  $p$  the condition  $(p^k, q) > 1$  means  $p \mid q$ , so the inner sum is over a subset of the prime divisors of  $q$  and is at most  $\sum_{p \mid q} \log p \leq \sum_{d \mid q} \Lambda(d) = \log q$ . There are at most  $\log z / \log 2$  values of  $k$ , which gives the bound.  $\square$

### 3.3 Standard identities

The following facts are classical and are quoted from Mathlib; we record them so that they can be cited by number.

**Lemma 3.8** (Hyperbola identity). *For arithmetic functions  $f, g$  with values in a semiring and every real  $x$ ,*

$$\sum_{n \leq x} (f * g)(n) = \sum_{k \leq x} f(k) \sum_{l \leq x/k} g(l).$$

*Proof.* Expand  $(f * g)(n) = \sum_{kl=n} f(k)g(l)$  and interchange the order of summation: the pairs  $(k, l)$  with  $kl \leq x$  are exactly those with  $k \leq x$  and  $l \leq x/k$ .  $\square$

**Lemma 3.9** (Möbius identities). *As arithmetic functions,  $\mu * 1 = \delta_1$  and  $\Lambda * 1 = \log$ ; that is,  $\sum_{d|n} \mu(d) = 1_{n=1}$  and  $\sum_{d|n} \Lambda(d) = \log n$  for  $n \geq 1$ . Moreover, for functions  $F, G$  on  $\mathbb{N}_{\geq 1}$ ,*

$$F(n) = \sum_{d|n} G(d) \text{ for all } n \geq 1 \quad \Longleftrightarrow \quad G(n) = \sum_{d|n} \mu(n/d) F(d) \text{ for all } n \geq 1.$$

*Proof.* These are in Mathlib.  $\square$

**Lemma 3.10** (Elementary facts about  $\varphi$ ). *For all  $a, b, n \in \mathbb{N}$ ,*

$$\sum_{d|n} \varphi(d) = n, \quad \varphi(a)\varphi(b) \leq \varphi(ab), \quad \varphi(ab) = \varphi(a)\varphi(b) \text{ if } (a, b) = 1,$$

*and for  $q \geq 1$  the unit group  $(\mathbb{Z}/q\mathbb{Z})^*$  has exactly  $\varphi(q)$  elements.*

*Proof.* These are in Mathlib.  $\square$

**Lemma 3.11** (Orthogonality of Dirichlet characters). *Let  $q \geq 1$ , let  $a \in (\mathbb{Z}/q\mathbb{Z})^*$  and let  $b \in \mathbb{Z}/q\mathbb{Z}$ . Then*

$$\sum_{\chi \bmod q} \chi(a)^{-1} \chi(b) = \begin{cases} \varphi(q) & \text{if } a = b, \\ 0 & \text{otherwise.} \end{cases}$$

*For a unit  $a$  one has  $\chi(a)^{-1} = \overline{\chi(a)}$ , since  $|\chi(a)| = 1$ .*

*Proof.* This is in Mathlib.  $\square$

### 3.4 Restriction and dilation of arithmetic functions

**Definition 3.12** (Restriction). For an arithmetic function  $f$  and a set  $S \subseteq \mathbb{N}$ , let  $f|_S$  denote the arithmetic function  $f|_S(n) := f(n)1_{n \in S}$ . For  $r \in \mathbb{N}$  we write  $f_r$  for the restriction of  $f$  to the integers coprime to  $r$ :

$$f_r(n) := f(n)1_{(n,r)=1} = \begin{cases} f(n) & \text{if } (r, n) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Note that  $f_1 = f$  and that  $f_0$  is supported on  $n = 1$  only, since  $(0, n) = n$ . The same notation is used for arbitrary functions  $f : \mathbb{N} \rightarrow R$  with  $R$  having a zero; when  $f$  is an arithmetic function the two readings agree.

**Lemma 3.13** (Restriction is a convolution homomorphism). *Let  $S \subseteq \mathbb{N}$  be multiplicatively saturated:  $ab \in S$  if and only if  $a \in S$  and  $b \in S$ . Then for all arithmetic functions  $f, g$  (with values in a semiring),*

$$(f * g)|_S = f|_S * g|_S, \quad \text{and} \quad (f - g)|_S = f|_S - g|_S$$

*for any  $S$  when the values lie in a ring. In particular, since  $(ab, r) = 1$  if and only if  $(a, r) = 1$  and  $(b, r) = 1$ , this applies to  $S = \{n : (n, r) = 1\}$ :  $(f * g)_r = f_r * g_r$  for every  $r \in \mathbb{N}$ .*

*Proof.* Linearity of restriction is clear. For the convolution, evaluate both sides at  $n$ : the left side is  $1_{n \in S} \sum_{ab=n} f(a)g(b)$  and the right side is  $\sum_{ab=n} 1_{a \in S} 1_{b \in S} f(a)g(b)$ . For each pair  $ab = n$  saturation gives  $1_{n \in S} = 1_{a \in S} 1_{b \in S}$ .  $\square$

**Definition 3.14** (Dilation). For  $e \geq 1$  and an arithmetic function  $f$ , the *dilation* of  $f$  by  $e$  is the arithmetic function

$$(\delta_e * f)(n) := 1_{e|n} f(n/e),$$

i.e. the Dirichlet convolution of  $f$  with the indicator  $\delta_e$  of the singleton  $\{e\}$ .

**Lemma 3.15** (Properties of dilation). *Let  $e \geq 1$  and let  $f, g$  be real arithmetic functions. Then*

$$f * (\delta_e * g) = \delta_e * (f * g) = (\delta_e * f) * g,$$

*and for every real  $x$ ,*

$$\sum_{k \leq x} |(\delta_e * f)(k)| \leq \sum_{k \leq x} |f(k)|.$$

*Proof.* The identities are instances of associativity and commutativity of Dirichlet convolution. Concretely, at  $n = em$  all three functions equal  $\sum_{ab=m} f(a)g(b)$  after the substitution  $b \mapsto eb$  (respectively  $a \mapsto ea$ ) in the divisor sum, and all three vanish when  $e \nmid n$ .

For the inequality, the non-zero terms on the left have  $k = em$  with  $m \leq x/e$ , and  $|(\delta_e * f)(em)| = |f(m)|$ ; reindexing by  $m$  gives  $\sum_{m \leq x/e} |f(m)|$ , which is a sub-sum of the right-hand side since  $x/e \leq x$  and the terms are non-negative.  $\square$

**Lemma 3.16** (Submultiplicativity of the  $\ell^1$  norm). *For real arithmetic functions  $f, g$  and every real  $x \geq 0$ ,*

$$\sum_{k \leq x} |(f * g)(k)| \leq \left( \sum_{k \leq x} |f(k)| \right) \left( \sum_{k \leq x} |g(k)| \right).$$

*Proof.* By the triangle inequality the left side is at most  $\sum_{k \leq x} \sum_{ab=k} |f(a)||g(b)|$ . The pairs  $(a, b)$  with  $ab = k \leq x$  for the various  $k$  are pairwise distinct and all satisfy  $1 \leq a \leq x$ ,  $1 \leq b \leq x$ , so this double sum is at most  $\sum_{a \leq x} \sum_{b \leq x} |f(a)||g(b)|$ .  $\square$

### 3.5 The discrepancy $\Delta_f$

**Definition 3.17** (The discrepancy  $\Delta_f$ ). Let  $f : \mathbb{N} \rightarrow R$  with  $R$  a field (in practice  $R = \mathbb{R}$  or  $\mathbb{C}$ ), let  $x \in \mathbb{R}$ ,  $q \in \mathbb{N}$  and  $a \in \mathbb{Z}/q\mathbb{Z}$ . The *discrepancy* of  $f$  in the progression  $a \bmod q$  up to  $x$  is

$$\Delta_f(x; q, a) := \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} f(n) - \frac{1}{\varphi(q)} \sum_{\substack{n \leq x \\ (n, q) = 1}} f(n).$$

Here  $1/\varphi(0)$  is read as 0, so that for  $q = 0$  only the first sum survives; all estimates below assume  $q \geq 1$ .

**Lemma 3.18** (Linearity of  $\Delta$ ). *For fixed  $x, q, a$  the map  $f \mapsto \Delta_f(x; q, a)$  is  $R$ -linear:  $\Delta_{f+g} = \Delta_f + \Delta_g$ ,  $\Delta_{f-g} = \Delta_f - \Delta_g$ ,  $\Delta_{cf} = c\Delta_f$  for  $c \in R$ , and  $\Delta_{\sum_{i \in I} f_i} = \sum_{i \in I} \Delta_{f_i}$  for every finite index set  $I$ .*

*Proof.* Both sums in Definition 3.17 are finite sums of values of  $f$ , hence linear in  $f$ . The finite-sum version follows from additivity by induction on  $I$ .  $\square$

**Lemma 3.19** (Trivial bound for  $\Delta_f$ ). *For every  $f : \mathbb{N} \rightarrow \mathbb{R}$ , every real  $z$ , every  $q \in \mathbb{N}$  and every  $a \in \mathbb{Z}/q\mathbb{Z}$ ,*

$$|\Delta_f(z; q, a)| \leq 2 \sum_{n \leq z} |f(n)|.$$

*If moreover  $f \geq 0$  and  $q \geq 1$ , then  $|\Delta_f(z; q, a)| \leq 2 \sum_{n \leq z} f(n)$ .*

*Proof.* Each of the two sums in Definition 3.17 is a sub-sum of  $\sum_{n \leq z} f(n)$ , so each is bounded in absolute value by  $\sum_{n \leq z} |f(n)|$ ; moreover  $1/\varphi(q) \leq 1$  (this also holds for  $q = 0$ , where the factor is 0). The triangle inequality gives the claim. When  $f \geq 0$  both sub-sums lie in  $[0, \sum_{n \leq z} f(n)]$ , so their difference (with the second scaled by  $1/\varphi(q) \in (0, 1]$ ) has absolute value at most  $2 \sum_{n \leq z} f(n)$ .  $\square$

**Lemma 3.20** (Restricting to integers coprime to the modulus). *Let  $f : \mathbb{N} \rightarrow R$ ,  $y \in \mathbb{R}$ ,  $q \in \mathbb{N}$  and  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ . Then*

$$\Delta_{f_q}(y; q, a) = \Delta_f(y; q, a),$$

*with  $f_q$  as in Definition 3.12.*

*Proof.* If  $n \equiv a \pmod{q}$  with  $a$  a unit, then  $n$  is a unit modulo  $q$ , i.e.  $(n, q) = 1$ , so  $f_q(n) = f(n)$  for every  $n$  in the first sum of Definition 3.17. On the second sum  $f_q = f$  by definition.  $\square$

**Lemma 3.21** ( $\Delta_\Lambda$  and the Chebyshev error). *For all  $z \in \mathbb{R}$ ,  $q \in \mathbb{N}$  and  $a \in \mathbb{Z}/q\mathbb{Z}$ ,*

$$\Delta_\Lambda(z; q, a) = \psi(z; q, a) - \frac{1}{\varphi(q)} \sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n),$$

*and consequently*

$$\psi(z; q, a) - \frac{z}{\varphi(q)} = \Delta_\Lambda(z; q, a) + \frac{1}{\varphi(q)} \left( \sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n) - z \right).$$

*Proof.* The first identity is the definition of  $\Delta_\Lambda$  together with the definition of  $\psi(z; q, a)$ ; the second follows by adding and subtracting  $z/\varphi(q)$ .  $\square$

**Lemma 3.22** ( $\Delta_f$  via Dirichlet characters). *Let  $f : \mathbb{N} \rightarrow \mathbb{R}$  or  $f : \mathbb{N} \rightarrow \mathbb{C}$ , let  $y \in \mathbb{R}$ ,  $q \geq 1$  and  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ . Then*

$$\Delta_f(y; q, a) = \frac{1}{\varphi(q)} \sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} \overline{\chi(a)} \sum_{n \leq y} f(n) \chi(n),$$

*where  $\chi_0$  is the principal character modulo  $q$ .*

*Proof.* By Lemma 3.11,  $\sum_{\chi \bmod q} \overline{\chi(a)} \chi(n) = \varphi(q) 1_{n \equiv a \pmod{q}}$  for every  $n$ , since  $a$  is a unit. Summing over all characters therefore gives  $\sum_{\chi} \overline{\chi(a)} \sum_{n \leq y} f(n) \chi(n) = \varphi(q) \sum_{n \leq y, n \equiv a} f(n)$ . The principal character contributes  $\sum_{n \leq y} f(n) \chi_0(n) = \sum_{n \leq y, (n, q) = 1} f(n)$ , because  $\chi_0(a) = 1$ . Subtracting the principal term and dividing by  $\varphi(q)$  gives exactly the two sums in Definition 3.17.  $\square$

### 3.6 The parameters and the maximal quantities

**Definition 3.23** (Parameter datum). A *parameter datum* consists of real numbers  $x, U, V$  satisfying

$$x \geq 2, \quad UV \leq \sqrt{x}, \quad U \geq e^{\sqrt{\log x}}, \quad V \geq e^{\sqrt{\log x}}.$$

Every statement in Chapters 6 to 10 that mentions  $x, U$  or  $V$  without quantifying them is implicitly universally quantified over such a datum, and the implied constants in those statements never depend on the datum. The main theorem is proved by instantiating  $U = V = e^{\sqrt{\log x}}$  once  $x > e^{32}$  (Theorem 3).

**Lemma 3.24** (Consequences of the parameter constraints). *For every parameter datum  $(x, U, V)$ ,*

$$\log x \geq 16, \quad x \geq e^{16}, \quad 2 \leq \sqrt{x} \leq x, \quad e \leq U \leq \sqrt{x} \leq x, \quad e \leq V \leq \sqrt{x} \leq x.$$

*Proof.* Since  $\sqrt{\log x} \geq 0$  we have  $U, V \geq e^{\sqrt{\log x}} \geq 1$ . Multiplying the two lower bounds and using  $UV \leq \sqrt{x}$  gives  $e^{2\sqrt{\log x}} \leq UV \leq \sqrt{x} = e^{(\log x)/2}$ , so  $2\sqrt{\log x} \leq \frac{1}{2} \log x$ . Writing  $s = \sqrt{\log x} > 0$  this reads  $4s \leq s^2$ , i.e.  $s \geq 4$ , so  $\log x \geq 16$  and  $x \geq e^{16}$ . In particular  $x \geq 4$ , so  $\sqrt{x} \geq 2$ , and  $\sqrt{x} \leq x$  as  $x \geq 1$ . Finally  $U \leq UV \leq \sqrt{x}$  because  $V \geq 1$ , and  $U \geq e^{\sqrt{\log x}} \geq e$  because  $\log x \geq 1$ ; symmetrically for  $V$ .  $\square$

**Definition 3.25** (Maxima over  $y$  and over reduced residues). Fix a parameter datum. For a function  $F$  on  $\mathbb{R}$  with values in  $[0, \infty]$  we write

$$\max_{\sqrt{x} \leq y \leq x} F(y) := \sup\{F(y) : \sqrt{x} \leq y \leq x\},$$

and for  $q \in \mathbb{N}$  and  $F$  a function on  $\mathbb{R} \times \mathbb{Z}/q\mathbb{Z}$  with values in  $[0, \infty]$ ,

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} F(y, a) := \sup_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \sup_{\sqrt{x} \leq y \leq x} F(y, a).$$

Both suprema are taken in  $[0, \infty]$ , so they always exist, and the residue  $a$  ranges over the units of  $\mathbb{Z}/q\mathbb{Z}$  only. We apply this to  $F(y, a) = |\Delta_f(y; q, a)|$ , and when  $y$  and  $a$  are clear from the context we abbreviate the left-hand side to  $\max_{y,a} |\Delta_f(y; q, a)|$ . It is part of every estimate for such a maximum that the supremum is finite.

**Lemma 3.26** (Working with the maxima). *Fix a parameter datum, let  $q \in \mathbb{N}$  and let  $f, g : \mathbb{N} \rightarrow \mathbb{R}$ .*

1. *If  $a \in (\mathbb{Z}/q\mathbb{Z})^*$  and  $\sqrt{x} \leq y \leq x$ , then  $|\Delta_f(y; q, a)| \leq \max_{y,a} |\Delta_f(y; q, a)|$ .*
2. *If  $|\Delta_f(y; q, a)| \leq B$  for all  $a \in (\mathbb{Z}/q\mathbb{Z})^*$  and all  $\sqrt{x} \leq y \leq x$ , then  $\max_{y,a} |\Delta_f(y; q, a)| \leq B$ ; in particular the maximum is always finite.*
3.  $\max_{y,a} |\Delta_{f+g}(y; q, a)| \leq \max_{y,a} |\Delta_f(y; q, a)| + \max_{y,a} |\Delta_g(y; q, a)|$ .

*The analogues of (1) and (2) hold for  $\max_{\sqrt{x} \leq y \leq x} F(y)$  without the residue variable.*

*Proof.* (1) and (2) are the defining properties of a supremum in the complete lattice  $[0, \infty]$ . Finiteness in (2) follows because, by Lemma 3.19,  $|\Delta_f(y; q, a)| \leq 2 \sum_{n \leq x} |f(n)|$  for all  $y \leq x$ , which is a real number. For (3), fix a unit  $a$  and  $\sqrt{x} \leq y \leq x$ . By Lemma 3.18 and the triangle inequality,  $|\Delta_{f+g}(y; q, a)| \leq |\Delta_f(y; q, a)| + |\Delta_g(y; q, a)|$ , and each term on the right is at most the corresponding maximum by (1). Taking the supremum over  $y$  and  $a$  on the left, using (2), gives the claim.  $\square$

# Chapter 4

## The two external inputs

Two analytic results are quoted from the literature and taken as axioms in Lean. Both are stated with an unspecified constant, which is itself part of the axiom, and in accordance with Notation 3.1 every implied constant in this document is allowed to depend on these two constants. A statement whose proof invokes one of the two inputs, directly or through an earlier result, lists it among the results its proof uses; in this sense every estimate from Corollary 4.1 onwards should be read as beginning with “Assume Theorem 1 and Theorem 2”.

### 4.1 The Siegel–Walfisz theorem

**Theorem 1** (Siegel–Walfisz). *Let  $A, C \in \mathbb{N}$ . For all real  $x \geq 2$ , all integers  $1 \leq q \leq (\log x)^C$  and all  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ ,*

$$\psi(x; q, a) = \frac{x}{\varphi(q)} + O_{A,C} \left( \frac{x}{(\log x)^A} \right).$$

*Proof.* This is the Siegel–Walfisz theorem [7, 11] in the form “an arbitrary power of log savings, uniformly for  $q \leq (\log x)^C$ ”; see for instance [2, Ch. 22] or [4, Thm. 13.5]. It is taken as an external input, and no proof is given here. The implied constant is ineffective. Only the cases  $C = 0$  (Corollary 4.1) and  $C$  even (Lemma 9.5) are used.  $\square$

**Corollary 4.1** (The prime number theorem with logarithmic savings). *Let  $A \in \mathbb{N}$ . For every real  $z \geq 2$ ,*

$$\sum_{n \leq z} \Lambda(n) = z + O_A \left( \frac{z}{(\log z)^A} \right),$$

*and for every integer  $q \geq 1$ ,*

$$\sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n) = z + O_A \left( \frac{z}{(\log z)^A} \right) + O(\log q \cdot \log z),$$

*the last error term having an absolute implied constant.*

*Proof.* For the first statement apply Theorem 1 with  $C = 0$ ,  $q = 1$  (note  $1 \leq (\log z)^0 = 1$ ) and  $a = 1$ : then  $\psi(z; 1, 1) = \sum_{n \leq z} \Lambda(n)$  and  $\varphi(1) = 1$ . For the second, write

$$\sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n) - z = \left( \sum_{n \leq z} \Lambda(n) - z \right) - \sum_{\substack{n \leq z \\ (n, q) > 1}} \Lambda(n)$$

and apply the triangle inequality: the first bracket is bounded by the first statement and the second sum by Lemma 3.7.  $\square$

## 4.2 The large sieve inequality

**Theorem 2** (The large sieve inequality). *For all real  $Q \geq 1$ , all  $H \in \mathbb{Z}$ , all integers  $N \geq 1$  and all  $c : \mathbb{Z} \rightarrow \mathbb{C}$ ,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \left| \sum_{H < n \leq H+N} c_n \chi(n) \right|^2 \ll (N + Q^2) \sum_{H < n \leq H+N} |c_n|^2,$$

where  $\sum^*$  denotes a sum over primitive characters and the implied constant is absolute.

*Proof.* This is the multiplicative form of the large sieve inequality, due to Bombieri and Davenport and in this sharp form to Montgomery and Vaughan [5]; see [3, Thm. 7.13] or [4, Thm. 25.6]. It is taken as an external input, and no proof is given here. (The inequality is known with the constant 1 in place of the implied constant; nothing below depends on this.)  $\square$

# Chapter 5

## Auxiliary estimates

This chapter collects the elementary estimates that the main argument needs: the average order of the divisor function, the convergent sums involving  $1/\varphi$ , and comparisons between powers of  $\log x$  and powers of  $x$ . None of them involves the external inputs.

### 5.1 The divisor function and Euler's totient

**Lemma 5.1** (The divisor function). *For every  $n \in \mathbb{N}$  we have  $\tau(n) \leq 2\sqrt{n}$ , and for every real  $Q \geq 2$ ,*

$$\sum_{q \leq Q} \tau(q) \ll Q \log Q$$

*with an absolute implied constant.*

*Proof.* For the first bound, the case  $n = 0$  is trivial (we take  $\tau(0) = 0$ ). For  $n \geq 1$  the map  $d \mapsto \min(d, n/d)$  sends the divisors of  $n$  into  $\{1, \dots, \lfloor \sqrt{n} \rfloor\}$ , and each value has at most two preimages  $\{d, n/d\}$ . Hence  $\tau(n) \leq 2\lfloor \sqrt{n} \rfloor \leq 2\sqrt{n}$ .

For the second, let  $N = \lfloor Q \rfloor$ . Counting the pairs  $(q, d)$  with  $d \mid q \leq N$  according to  $d$  gives  $\sum_{q \leq N} \tau(q) = \sum_{d \leq N} \lfloor N/d \rfloor \leq N \sum_{d \leq N} 1/d$ . The harmonic sum is at most  $1 + \log N \leq 1 + \log Q \ll \log Q$ , since  $Q \geq 2$ .  $\square$

**Lemma 5.2** (A lower bound for  $\varphi$ ). *For every  $d \in \mathbb{N}$ ,  $d \leq 2\varphi(d)^2$ . Consequently  $\frac{1}{d\varphi(d)} \leq \sqrt{2}d^{-3/2}$  for  $d \geq 1$ , the series  $\sum_{d \geq 1} \frac{1}{d\varphi(d)}$  converges, and for every real  $Q$ ,*

$$\sum_{r \leq Q} \frac{1}{r\varphi(r)} \ll 1.$$

*Proof.* The case  $d = 0$  is trivial. For  $d \geq 1$  both  $d$  and  $\varphi(d)^2$  are multiplicative (Lemma 3.10), so it suffices to compare them on prime powers and to check that the factor 2 is needed at most once. For  $p^k$  with  $k \geq 1$  one has  $\varphi(p^k)^2 = p^{2k-2}(p-1)^2 \geq p^k$  unless  $p = 2$ , in which case  $\varphi(2^k)^2 = 2^{2k-2} \geq 2^k/2$ . Multiplying over the prime factorisation of  $d$ , in which the prime 2 appears at most once, gives  $d \leq 2\varphi(d)^2$ .

Hence  $\varphi(d) \geq \sqrt{d/2}$  and  $1/(d\varphi(d)) \leq \sqrt{2}d^{-3/2}$ , so the series converges by comparison with  $\sum d^{-3/2}$ . A partial sum of a series with non-negative terms is at most its sum.  $\square$

**Lemma 5.3** (The sum of  $1/\varphi$ ). *Let  $f_\varphi(d) := \mu(d)^2/(d\varphi(d))$ . Then, as arithmetic functions,*

$$\frac{1}{\varphi} = f_\varphi * \frac{1}{\text{id}}, \quad \text{i.e.} \quad \frac{1}{\varphi(n)} = \sum_{d|n} \frac{f_\varphi(d)}{d} \quad (n \geq 1),$$

where  $f_\varphi \geq 0$ ,  $f_\varphi(d) \leq \sqrt{2}d^{-3/2}$  and  $\sum_{d \geq 1} f_\varphi(d)$  converges. Consequently, for every parameter datum and every real  $Q \leq x$ ,

$$\sum_{n \leq Q} \frac{1}{\varphi(n)} \ll \log x,$$

with an absolute implied constant.

*Proof. Step 1: the convolution identity.* All three functions  $1/\varphi$ ,  $f_\varphi$  and  $1/\text{id}$  are multiplicative (Lemma 3.10), so it suffices to compare both sides at prime powers  $p^k$ ,  $k \geq 1$ . Since  $\mu(p^j)^2 = 0$  for  $j \geq 2$ ,

$$\left(f_\varphi * \frac{1}{\text{id}}\right)(p^k) = \frac{1}{p^k} + \frac{1}{p(p-1)} \cdot \frac{1}{p^{k-1}} = \frac{1}{p^{k-1}} \cdot \frac{p-1+1}{p(p-1)} = \frac{1}{p^{k-1}(p-1)} = \frac{1}{\varphi(p^k)}.$$

The bound  $f_\varphi(d) \leq 1/(d\varphi(d)) \leq \sqrt{2}d^{-3/2}$  is Lemma 5.2, and it gives the convergence.

*Step 2: the sum.* If  $Q < 1$  the sum is empty. Otherwise, by Step 1 and the hyperbola identity (Lemma 3.8),

$$\sum_{n \leq Q} \frac{1}{\varphi(n)} = \sum_{d \leq Q} f_\varphi(d) \sum_{e \leq Q/d} \frac{1}{e}.$$

Each inner sum is a harmonic number, bounded by  $1 + \log(Q/d) \leq 1 + \log x \leq 2 \log x$  since  $\log x \geq 1$  (Lemma 3.24), and  $\sum_{d \leq Q} f_\varphi(d) \leq \sum_{d \geq 1} f_\varphi(d) < \infty$  because  $f_\varphi \geq 0$ .  $\square$

## 5.2 Powers of logarithms

**Lemma 5.4** (Powers of log against powers of  $x$ ).

1. For every real  $x \geq 1$ , every  $M \in \mathbb{N}$  and every real  $\delta > 0$ ,  $(\log x)^M \ll_{M,\delta} x^\delta$ .
2. Let  $B \in \mathbb{N}$  and let  $1 < z \leq x$  be real numbers with  $\sqrt{x} \leq z$ . Then  $\frac{z}{(\log z)^B} \leq 2^B \frac{x}{(\log x)^B}$ .
3. For every parameter datum, every real  $W \geq e^{\sqrt{\log x}}$  and every  $n \in \mathbb{N}$ ,  $(\log x)^n \ll_n \sqrt{W}$ . In particular this applies to  $W = U$  and  $W = V$ .

*Proof.* (1) For  $M = 0$  both sides equal 1. For  $M \geq 1$  put  $c = \delta/M$ . Then  $\log x = c^{-1} \log(x^c) \leq c^{-1}(x^c - 1) \leq c^{-1}x^c$ , using  $\log t \leq t - 1$  for  $t > 0$ . Raising to the  $M$ -th power gives  $(\log x)^M \leq c^{-M}x^{cM} = (M/\delta)^M x^\delta$ .

(2) From  $z \geq \sqrt{x}$  we get  $\log z \geq \frac{1}{2} \log x > 0$ , so  $(\log x)^B \leq 2^B (\log z)^B$ . Combining with  $z \leq x$  gives  $z(\log x)^B \leq 2^B x(\log z)^B$ , which is the claim after dividing by the positive quantity  $(\log z)^B$ .

(3) Put  $s = \sqrt{\log x} \geq 0$ , so  $(\log x)^n = s^{2n}$ . From the Taylor series of the exponential,  $(s/2)^{2n}/(2n)! \leq e^{s/2}$ , i.e.  $s^{2n} \leq 2^{2n}(2n)! e^{s/2}$ , and  $e^{s/2} = \sqrt{e^s} \leq \sqrt{W}$ .  $\square$

**Lemma 5.5** (Absorbing powers of  $\log x$  into  $x/(\log x)^N$ ). *For every parameter datum, all  $j, N \in \mathbb{N}$  and every integer  $q \leq \sqrt{x}$ ,*

$$(\log x)^j \ll_{j,N} \frac{x}{(\log x)^N}, \quad \sqrt{x} (\log x)^j \ll_{j,N} \frac{x}{(\log x)^N}, \quad \tau(q) \sqrt{x} \log x \ll_N \frac{x}{(\log x)^N}.$$

*Proof.* By Lemma 5.4(1) with  $M = j + N$  and  $\delta = 1/2$ ,  $(\log x)^{j+N} \ll_{j,N} \sqrt{x} \leq x$ , which gives the first two bounds after dividing by  $(\log x)^N$  (for the second use  $\sqrt{x} \cdot \sqrt{x} = x$ ). For the third,  $\tau(q) \leq 2\sqrt{q} \leq 2x^{1/4}$  by Lemma 5.1, and Lemma 5.4(1) with  $M = N + 1$ ,  $\delta = 1/4$  gives  $(\log x)^{N+1} \ll_N x^{1/4}$ ; then  $x^{1/4} \cdot \sqrt{x} \cdot x^{1/4} = x$ .  $\square$

## Chapter 6

# Vaughan's identity

Throughout the rest of the blueprint a parameter datum  $(x, U, V)$  (Definition 3.23) is fixed. Vaughan's identity [8] writes  $\Lambda$  as a sum of three pieces, each of which is amenable to a different technique. We use it in the form of [4, §26.1].

**Definition 6.1** (The pieces of Vaughan's identity). Let  $\Lambda_{\leq U} := \Lambda \cdot 1_{[1, U]}$  and  $\mu_{\leq V} := \mu \cdot 1_{[1, V]}$  be the truncated von Mangoldt and Möbius functions, so that  $\Lambda_{\leq U}(n) = \Lambda(n)$  if  $n \leq U$  and  $\Lambda_{\leq U}(n) = 0$  if  $n > U$ , and similarly for  $\mu_{\leq V}$ ; write  $\Lambda_{> U} := \Lambda - \Lambda_{\leq U}$  and  $\mu_{> V} := \mu - \mu_{\leq V}$ . The *Type I part* and the *Type II part* of  $\Lambda$  are the arithmetic functions

$$\Lambda^{\sharp} := \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1, \quad \Lambda^{\flat} := (\Lambda_{> U} * 1) * \mu_{> V}.$$

**Proposition 6.2** (Vaughan's identity). For every  $n \in \mathbb{N}$ ,

$$\Lambda(n) = \Lambda^{\sharp}(n) + \Lambda^{\flat}(n) + \Lambda_{\leq U}(n).$$

*Proof.* Expand  $\Lambda^{\flat}$  by bilinearity of the Dirichlet convolution and use the identities  $\Lambda * 1 = \log$  and  $1 * \mu = \delta_1$  of Lemma 3.9:

$$\Lambda^{\flat} = \Lambda * 1 * \mu - \Lambda * 1 * \mu_{\leq V} - \Lambda_{\leq U} * 1 * \mu + \Lambda_{\leq U} * 1 * \mu_{\leq V} = \Lambda - \log * \mu_{\leq V} - \Lambda_{\leq U} + \Lambda_{\leq U} * \mu_{\leq V} * 1.$$

Adding  $\Lambda^{\sharp} = \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1$  and  $\Lambda_{\leq U}$ , everything cancels except  $\Lambda$ .  $\square$

**Lemma 6.3** (Splitting the maximal discrepancy). For every  $q \in \mathbb{N}$ ,

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda}(y; q, a)| \leq \max_{y, a} |\Delta_{\Lambda^{\sharp}}(y; q, a)| + \max_{y, a} |\Delta_{\Lambda^{\flat}}(y; q, a)| + \max_{y, a} |\Delta_{\Lambda_{\leq U}}(y; q, a)|.$$

*Proof.* By Proposition 6.2,  $\Lambda = (\Lambda^{\sharp} + \Lambda^{\flat}) + \Lambda_{\leq U}$  as functions on  $\mathbb{N}$ . Apply the triangle inequality for maximal discrepancies, Lemma 3.26(3), twice.  $\square$

## Chapter 7

# The small part $\Lambda_{\leq U}$

The contribution of  $\Lambda_{\leq U}$  is bounded trivially: the function is non-negative and supported on  $[1, U]$  with  $U \leq \sqrt{x}$ , and there are at most  $Q \leq \sqrt{x}/(\log x)^{A+3}$  moduli. Along the way we record the  $\ell^1$  norms of the two coefficient functions  $\Lambda_{\leq U}$  and  $\mu_{\leq V}$ , which are also needed in the Type I analysis.

**Lemma 7.1** ( $\ell^1$  norms of the coefficient functions). *For every real  $y$ ,*

$$\sum_{n \leq y} \Lambda_{\leq U}(n) \leq U \log x \quad \text{and} \quad \sum_{n \leq y} |\mu_{\leq V}(n)| \leq V;$$

*moreover  $0 \leq \Lambda_{\leq U}(n) \leq \log n$  for every  $n \geq 1$ .*

*Proof.* The function  $\Lambda_{\leq U}$  is non-negative and supported on the at most  $U$  integers  $1 \leq n \leq U$ , and on its support  $\Lambda_{\leq U}(n) = \Lambda(n) \leq \log n \leq \log U \leq \log x$  by Lemma 3.24. Hence its sum over any range is at most  $U \log x$ . Likewise  $\mu_{\leq V}$  is supported on the at most  $V$  integers  $1 \leq n \leq V$  and satisfies  $|\mu_{\leq V}| \leq 1$ .  $\square$

**Lemma 7.2** (Pointwise bound for  $\Delta_{\Lambda_{\leq U}}$ ). *For every real  $y$ , every  $q \geq 1$ , every  $a \in \mathbb{Z}/q\mathbb{Z}$  and every  $r \in \mathbb{N}$ ,*

$$|\Delta_{\Lambda_{\leq U}}(y; q, a)| \leq 2U \log x \quad \text{and} \quad |\Delta_{(\Lambda_{\leq U})_r}(y; q, a)| \leq 2U \log x.$$

*Proof.* Both  $\Lambda_{\leq U}$  and its restriction  $(\Lambda_{\leq U})_r$  are non-negative and bounded pointwise by  $\Lambda_{\leq U}$ , so by Lemma 3.19 their discrepancies are bounded in absolute value by  $2 \sum_{n \leq y} \Lambda_{\leq U}(n) \leq 2U \log x$  (Lemma 7.1).  $\square$

**Proposition 7.3** (The small part on average). *For every  $A \in \mathbb{N}$  and every real  $0 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda_{\leq U}}(y; q, a)| \leq \frac{2x}{(\log x)^{A+2}}.$$

*Proof.* By Lemma 7.2 and Lemma 3.26(2), each of the  $\lfloor Q \rfloor \leq Q$  summands is at most  $2U \log x$ . Using  $U \leq \sqrt{x}$  (Lemma 3.24) and the hypothesis on  $Q$ ,

$$Q \cdot 2U \log x \leq \frac{\sqrt{x}}{(\log x)^{A+3}} \cdot 2\sqrt{x} \log x = \frac{2x}{(\log x)^{A+2}}. \quad \square$$

# Chapter 8

## Type I sums: $\Lambda^\sharp$

The Type I part  $\Lambda^\sharp = \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1$  is a convolution of a coefficient function of small support ( $\mu_{\leq V}$ , respectively  $\Lambda_{\leq U} * \mu_{\leq V}$ ) with one of the smooth functions  $\log$  and  $1$ . The discrepancy of such a convolution is a short sum of discrepancies of the smooth factor at the points  $y/k$  (Lemma 8.1), and those are bounded pointwise, uniformly in the modulus, by Abel summation against the trivial bound  $|\Delta_1| \leq 1$ . This gives a pointwise bound of size  $UV \log x$  for  $\Delta_{\Lambda^\sharp}(y; q, a)$ , which is summed trivially over  $q \leq Q$ .

There is one complication. What the conductor analysis of Chapter 9 produces is not  $\Lambda^\sharp$  but its restriction  $\Lambda_r^\sharp$  to integers coprime to  $r$ , and restriction turns the smooth factor  $\log$  into  $\log_r$ , which is neither smooth nor monotone. We therefore prove the pointwise bound for  $\Lambda_r^\sharp$ , expanding  $\log_r$  and  $1_r$  as Möbius sums of dilations of  $\log$  and  $1$ ; this costs a factor  $\tau(r)$ , which is harmless in both applications.

### 8.1 Discrepancies of smooth weights

**Lemma 8.1** ( $\Delta$  of a Dirichlet convolution). *Let  $f, g$  be arithmetic functions with values in  $\mathbb{R}$  or  $\mathbb{C}$ , let  $y \in \mathbb{R}$ ,  $q \geq 1$  and  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ . Then*

$$\Delta_{f*g}(y; q, a) = \sum_{\substack{k \leq y \\ (k, q) = 1}} f(k) \Delta_g\left(\frac{y}{k}; q, a\bar{k}\right),$$

where  $\bar{k}$  denotes the inverse of  $k$  in  $(\mathbb{Z}/q\mathbb{Z})^*$ .

*Proof.* Both sides may be computed in  $\mathbb{C}$ . By Lemma 3.22,  $\varphi(q)\Delta_{f*g}(y; q, a) = \sum_{\chi \neq \chi_0} \overline{\chi(a)} \sum_{n \leq y} (f * g)(n)\chi(n)$ . Since  $\chi$  is completely multiplicative,  $(f * g)(n)\chi(n) = ((f\chi) * (g\chi))(n)$ , and the hyperbola identity (Lemma 3.8) gives

$$\varphi(q)\Delta_{f*g}(y; q, a) = \sum_{k \leq y} f(k) \sum_{\chi \neq \chi_0} \overline{\chi(a)} \chi(k) \sum_{l \leq y/k} g(l)\chi(l).$$

If  $(k, q) > 1$  then  $\chi(k) = 0$  and the term vanishes. If  $(k, q) = 1$  then  $k$  is a unit and  $\overline{\chi(a)}\chi(k) = \chi(a\bar{k})$ , so by Lemma 3.22 applied to  $g$  at the point  $y/k$  and the unit  $a\bar{k}$ , the inner double sum equals  $\varphi(q)\Delta_g(y/k; q, a\bar{k})$ . Dividing by  $\varphi(q)$  gives the claim.  $\square$

**Lemma 8.2** (The discrepancy of the constant function 1). *For every real  $x$ , every  $q \geq 1$  and every  $a \in \mathbb{Z}/q\mathbb{Z}$  (not necessarily a unit),*

$$|\Delta_1(x; q, a)| \leq 1.$$

*Proof.* Let  $N = \lfloor x \rfloor$  and write  $N = kq + m$  with  $0 \leq m < q$ . The integers  $1 \leq n \leq N$  form  $k$  complete residue systems modulo  $q$  followed by  $m$  further integers. Each complete system contains exactly one  $n \equiv a \pmod{q}$  and exactly  $\varphi(q)$  integers coprime to  $q$  (Lemma 3.10). Hence

$$\#\{n \leq N : n \equiv a\} = k + c, \quad \#\{n \leq N : (n, q) = 1\} = \varphi(q)k + c',$$

with  $0 \leq c \leq 1$  and  $0 \leq c' \leq \varphi(q)$  counting the contributions of the last  $m$  integers. Therefore  $\Delta_1(x; q, a) = c - c'/\varphi(q) \in [-1, 1]$ .  $\square$

**Lemma 8.3** (Abel summation for  $\Delta$ ). *Let  $q \geq 1$ ,  $a \in (\mathbb{Z}/q\mathbb{Z})^*$  and  $x \geq 1$ . Let  $g, g' : \mathbb{R} \rightarrow \mathbb{C}$  be such that  $g$  has derivative  $g'(t)$  at every  $t \in [1, x]$ ,  $g'$  is locally integrable on  $[1, x]$ , and  $g(1) = 0$ . Then*

$$\Delta_g(x; q, a) = \Delta_1(x; q, a)g(x) - \int_1^x \Delta_1(t; q, a)g'(t)dt,$$

where  $\Delta_g$  is the discrepancy of the arithmetic function  $n \mapsto g(n)$ . The same holds for real-valued  $g, g'$ .

*Proof.* By Lemma 3.22 both sides are, up to the factor  $1/\varphi(q)$ , sums over the non-principal characters  $\chi \pmod{q}$  of  $\overline{\chi(a)}$  times a character sum, so it suffices to prove the identity character by character; the integral may be exchanged with the finite sum because each integrand is the product of the locally integrable  $g'$  with a locally bounded step function. For a fixed  $\chi$  put  $S(t) = \sum_{n \leq t} \chi(n)$ . Abel's summation formula gives

$$\sum_{1 < n \leq x} g(n)\chi(n) = g(x)S(x) - g(1)S(1) - \int_1^x g'(t)S(t)dt,$$

and the term  $n = 1$  as well as the boundary term at 1 vanish because  $g(1) = 0$ . Recombining over  $\chi$  gives the identity with  $\Delta_1$  in place of  $S$ . The real case follows by viewing  $g, g'$  as complex-valued.  $\square$

**Lemma 8.4** (Discrepancy of a monotone weight). *Let  $q \geq 1$ ,  $a \in (\mathbb{Z}/q\mathbb{Z})^*$  and  $x \geq 1$ . Let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable at every point of  $[1, x]$ , with  $g'$  locally integrable on  $[1, x]$ , monotone increasing on  $[1, x]$  and  $g(1) = 0$ . Then*

$$|\Delta_g(x; q, a)| \leq 2g(x).$$

*Proof.* Apply Lemma 8.3 and bound the two terms. By Lemma 8.2,  $|\Delta_1(x; q, a)g(x)| \leq g(x)$ , where  $g(x) \geq g(1) = 0$  by monotonicity. Since  $g$  is monotone on  $[1, x]$ , its derivative satisfies  $g' \geq 0$  on  $(1, x)$ , so

$$\left| \int_1^x \Delta_1(t; q, a)g'(t)dt \right| \leq \int_1^x g'(t)dt = g(x) - g(1) = g(x)$$

by the fundamental theorem of calculus. Adding the two bounds gives  $2g(x)$ .  $\square$

**Proposition 8.5** (Basic Type I estimate). *Let  $v \in \mathbb{N}$  and let  $f$  be a real arithmetic function. For every real  $x \geq 2$ , every  $q \geq 1$  and every  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ ,*

$$|\Delta_{f * \log^v}(x; q, a)| \leq 2(\log x)^v \sum_{k \leq x} |f(k)|.$$

*In particular, for every  $e \geq 1$  and every real arithmetic function  $h$ ,*

$$|\Delta_{(\delta_e * h) * \log^v}(x; q, a)| \leq 2(\log x)^v \sum_{k \leq x} |h(k)|.$$

*Proof.* By Lemma 8.1,  $\Delta_{f * \log^v}(x; q, a) = \sum_{k \leq x, (k, q)=1} f(k) \Delta_{\log^v}(x/k; q, a\bar{k})$ , so by the triangle inequality it suffices to show  $|\Delta_{\log^v}(x/k; q, b)| \leq 2(\log x)^v$  for every unit  $b$  and every  $1 \leq k \leq x$ . If  $v = 0$  this is Lemma 8.2. If  $v \geq 1$ , apply Lemma 8.4 with  $g(t) = (\log t)^v$  on  $[1, x/k]$  (note  $x/k \geq 1$ ):  $g$  is differentiable with locally integrable derivative  $v(\log t)^{v-1}/t$ , monotone on  $[1, x/k]$  and  $g(1) = 0$ , so  $|\Delta_{\log^v}(x/k; q, b)| \leq 2(\log(x/k))^v \leq 2(\log x)^v$ .

The second statement is the first applied to  $f = \delta_e * h$ , together with  $\sum_{k \leq x} |(\delta_e * h)(k)| \leq \sum_{k \leq x} |h(k)|$  (Lemma 3.15).  $\square$

## 8.2 Möbius expansion of restricted smooth weights

**Lemma 8.6** (Möbius expansion of  $1_r$  and  $\log_r$ ). *Let  $r \geq 1$ . For every  $b \in \mathbb{N}$ ,*

$$\sum_{e|r} \mu(e) 1_{e|b} = 1_{(r,b)=1},$$

*and consequently, as arithmetic functions,*

$$1_r = \sum_{e|r} \mu(e) (\delta_e * 1), \quad \log_r = \sum_{e|r} \mu(e) (\log e \cdot (\delta_e * 1) + \delta_e * \log),$$

*where  $1_r$  and  $\log_r$  are the restrictions (Definition 3.12) of 1 and  $\log$  to the integers coprime to  $r$ . Convolution with an arbitrary real arithmetic function  $H$ ,*

$$H * 1_r = \sum_{e|r} \mu(e) ((\delta_e * H) * 1), \quad H * \log_r = \sum_{e|r} \mu(e) (\log e \cdot ((\delta_e * H) * 1) + (\delta_e * H) * \log).$$

*The identities for  $\log_r$  and  $H * \log_r$  hold for  $r = 0$  as well.*

*Proof. Step 1: the indicator.* The divisors  $e$  of  $r$  with  $e | b$  are exactly the divisors of  $\gcd(r, b)$ , and  $\sum_{e|m} \mu(e) = 1_{m=1}$  for  $m \geq 1$  (Lemma 3.9). Note that  $r \geq 1$  is needed here: for  $r = 0$  the left side is an empty sum while  $(0, 1) = 1$ .

*Step 2:  $1_r$  and  $\log_r$ .* Evaluate at  $b \geq 1$ . Since  $(\delta_e * 1)(b) = 1_{e|b}$ , the identity for  $1_r$  is Step 1. For  $\log_r$ ,  $(\delta_e * \log)(b) = 1_{e|b} \log(b/e)$ , so  $\log e \cdot (\delta_e * 1)(b) + (\delta_e * \log)(b) = 1_{e|b} (\log e + \log(b/e)) = 1_{e|b} \log b$ ; summing against  $\mu(e)$  over  $e | r$  and using Step 1 gives  $1_{(r,b)=1} \log b = \log_r(b)$ . At  $b = 0$  all terms vanish. For  $r = 0$  both sides of the  $\log_r$  identity vanish except at  $b = 1$ , where  $\log 1 = 0$ .

*Step 3: convolution with  $H$ .* Convolve the identities of Step 2 with  $H$ , using bilinearity of the convolution and  $H * (\delta_e * g) = (\delta_e * H) * g$  (Lemma 3.15).  $\square$

### 8.3 The pointwise bound for $\Lambda_r^\sharp$

**Proposition 8.7** (Pointwise Type I bound for  $\Lambda_r^\sharp$ ). *For every  $q \geq 1$ , every  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ , every  $r \in \mathbb{N}$  with  $r \leq x$  and every real  $2 \leq y \leq x$ ,*

$$|\Delta_{(\mu_{\leq V} * \log)_r}(y; q, a)| \leq 4\tau(r)V \log x, \quad |\Delta_{(\Lambda_{\leq U} * \mu_{\leq V} * 1)_r}(y; q, a)| \leq 2\tau(r)UV \log x,$$

and consequently

$$|\Delta_{\Lambda_r^\sharp}(y; q, a)| \ll \tau(r)UV \log x,$$

with an absolute implied constant.

*Proof. Step 1: the first term.* By Lemma 3.13,  $(\mu_{\leq V} * \log)_r = (\mu_{\leq V})_r * \log_r$ , and by Lemma 8.6 with  $H = (\mu_{\leq V})_r$ ,

$$(\mu_{\leq V} * \log)_r = \sum_{e|r} \mu(e) \left( \log e \cdot (\delta_e * H) * 1 + (\delta_e * H) * \log \right).$$

(If  $r = 0$  both sides are supported on  $n = 1$ , where  $\log 1 = 0$ , so the identity is trivial.) By linearity (Lemma 3.18) and the triangle inequality it suffices to bound each of the  $\tau(r)$  summands by  $4V \log x$ . Fix  $e | r$ ; then  $1 \leq e \leq r \leq x$ , so  $0 \leq \log e \leq \log x$ , and  $|\mu(e)| \leq 1$ . By Proposition 8.5 with  $v = 0$  and  $v = 1$  at the point  $y \geq 2$ ,

$$|\Delta_{(\delta_e * H) * 1}(y; q, a)| \leq 2\|H\|_1, \quad |\Delta_{(\delta_e * H) * \log}(y; q, a)| \leq 2 \log y \|H\|_1,$$

where  $\|H\|_1 = \sum_{k \leq y} |H(k)| \leq \sum_{k \leq y} |\mu_{\leq V}(k)| \leq V$  by Lemma 7.1. Since  $\log y \leq \log x$ , the  $e$ -th summand is at most  $\log x \cdot 2V + 2 \log x \cdot V = 4V \log x$ .

*Step 2: the second term.* If  $r = 0$ , the restricted function is supported on  $n = 1$ , where  $(\Lambda_{\leq U} * \mu_{\leq V} * 1)(1) = \Lambda(1)\mu(1) = 0$ , so the discrepancy vanishes. Let  $r \geq 1$ . By Lemma 3.13 and Lemma 8.6 with  $H = (\Lambda_{\leq U} * \mu_{\leq V})_r$ ,

$$(\Lambda_{\leq U} * \mu_{\leq V} * 1)_r = H * 1_r = \sum_{e|r} \mu(e) (\delta_e * H) * 1.$$

By Lemma 3.18, the triangle inequality,  $|\mu(e)| \leq 1$  and Proposition 8.5 with  $v = 0$ , each of the  $\tau(r)$  summands has discrepancy at most  $2\|H\|_1$ , where

$$\|H\|_1 \leq \sum_{k \leq y} |(\Lambda_{\leq U} * \mu_{\leq V})(k)| \leq \left( \sum_{k \leq y} \Lambda_{\leq U}(k) \right) \left( \sum_{k \leq y} |\mu_{\leq V}(k)| \right) \leq U \log x \cdot V$$

by Lemma 3.16 and Lemma 7.1.

*Step 3: conclusion.* Restriction is linear, so  $\Lambda_r^\sharp = (\mu_{\leq V} * \log)_r - (\Lambda_{\leq U} * \mu_{\leq V} * 1)_r$ , and by Lemma 3.18 and the triangle inequality  $|\Delta_{\Lambda_r^\sharp}(y; q, a)| \leq 4\tau(r)V \log x + 2\tau(r)UV \log x \leq 6\tau(r)UV \log x$ , using  $U \geq 1$  (Lemma 3.24).  $\square$

*Remark 8.8.* The factor  $\tau(r)$  cannot be dropped from Proposition 8.7: the discrepancy of  $\log_r$  is a signed combination of  $2^{\omega(r)}$  residue-class counts with no cancellation in general, and the Möbius expansion records this honestly. Downstream it is harmless, because  $\tau$  is only ever *averaged* over the modulus (Proposition 8.9) or bounded by  $2x^{1/4}$  (Lemma 9.6).

## 8.4 Averaging over the modulus

**Proposition 8.9** (The Type I part on average). *For every  $A \in \mathbb{N}$  and every real  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^\sharp}(y; q, a)| \ll \frac{x}{(\log x)^A},$$

*with an absolute implied constant.*

*Proof.* Since  $\log x \geq 1$ , the hypothesis gives  $Q \leq \sqrt{x} \leq x$ ; also  $\sqrt{x} \geq 2$  (Lemma 3.24), so every  $y \in [\sqrt{x}, x]$  satisfies  $2 \leq y \leq x$ .

*Step 1: the pointwise bound.* Fix  $1 \leq q \leq Q$ , a unit  $a$  and  $\sqrt{x} \leq y \leq x$ . By Lemma 3.20,  $\Delta_{\Lambda^\sharp}(y; q, a) = \Delta_{\Lambda_q^\sharp}(y; q, a)$ , and Proposition 8.7 with  $r = q \leq x$  gives  $|\Delta_{\Lambda^\sharp}(y; q, a)| \ll \tau(q) UV \log x$ . By Lemma 3.26(2) the same bound holds for the maximum over  $y$  and  $a$ .

*Step 2: averaging the divisor function.* Summing over  $q \leq Q$ ,

$$\sum_{q \leq Q} \max_{y, a} |\Delta_{\Lambda^\sharp}(y; q, a)| \ll UV \log x \sum_{q \leq Q} \tau(q) \ll UV \log x \cdot Q \log x,$$

where for  $Q \geq 2$  we used Lemma 5.1 and  $\log Q \leq \log x$ , while for  $1 \leq Q < 2$  the sum is  $\tau(1) = 1 \leq Q \log x$ .

*Step 3: conclusion.* Using  $UV \leq \sqrt{x}$  and  $Q(\log x)^{A+2} \leq Q(\log x)^{A+3} \leq \sqrt{x}$ ,

$$UV \cdot Q(\log x)^2 \cdot (\log x)^A = UV \cdot Q(\log x)^{A+2} \leq \sqrt{x} \cdot \sqrt{x} = x,$$

so the right-hand side of Step 2 is  $\ll x/(\log x)^A$ . □

## Chapter 9

# Type II sums: $\Lambda^b$

The Type II part  $\Lambda^b = (\Lambda_{>U} * 1) * \mu_{>V}$  is a convolution of two functions neither of which is smooth, and it cannot be bounded pointwise. Instead it is bounded on average over the modulus. By orthogonality,  $\varphi(q)\Delta_{\Lambda^b}(y; q, a)$  is a sum over the non-principal characters  $\chi \bmod q$  of character sums  $\sum_{n \leq y} \Lambda^b(n)\chi(n)$ . Grouping the characters by their conductor  $d \mid q$  produces, for each  $d$ , a sum over the primitive characters  $\xi$  modulo  $d$  of the twisted sums  $S_{q/d}(y, \xi) = |\sum_{n \leq y} \Lambda_{q/d}^b(n)\xi(n)|$ . The conductors  $d \leq (\log x)^C$  are treated by Siegel–Walfisz, after converting the primitive-character sums back into discrepancies by Möbius inversion (Section 9.2). The conductors  $d > (\log x)^C$  are collected, after the sum over  $q$  has been reorganised, into the quantities  $T_r(Q)$  of Section 9.3, which are bounded by a maximal large sieve inequality for Dirichlet convolutions (Section 9.4), applied to a dyadic decomposition of  $\Lambda^b$  (Section 9.5).

### 9.1 Characters and conductors

**Lemma 9.1** (Grouping characters by conductor). *Let  $q \geq 1$  and let  $F$  be a function on the Dirichlet characters modulo  $q$  with values in an additive commutative monoid. Then*

$$\sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} F(\chi) = \sum_{\substack{d \mid q \\ d \neq 1}} \sum_{\xi \bmod d}^* F(\xi \uparrow^q).$$

*Proof.* Group the characters  $\chi \bmod q$  according to their conductor  $d$ , which divides  $q$ . For a fixed  $d \mid q$ , the map  $\xi \mapsto \xi \uparrow^q$  is a bijection from the primitive characters modulo  $d$  onto the characters modulo  $q$  of conductor  $d$ : it is injective, it preserves the conductor, and every character of conductor  $d$  is induced by its (primitive) associated character modulo  $d$ . Finally a character modulo  $q$  is principal if and only if its conductor is 1, which accounts for the exclusion of  $d = 1$  on the right and of  $\chi_0$  on the left.  $\square$

**Lemma 9.2** (Primitive character sums via Möbius inversion). *Let  $q \geq 1$ ,  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ ,  $x \in \mathbb{R}$ , and let  $g : \mathbb{N} \rightarrow \mathbb{R}$  vanish at every  $n$  with  $(n, q) > 1$ . For  $i \geq 1$  put*

$$G_i := \sum_{\xi \bmod i}^* \overline{\xi(a)} \sum_{n \leq x} \xi(n)g(n),$$

where  $a$  is reduced modulo  $i$ . Then for every divisor  $d$  of  $q$ ,

$$\varphi(d) \Delta_g(x; d, a) = \sum_{\substack{i|d \\ i \neq 1}} G_i.$$

Consequently, for every real arithmetic function  $f$ , every  $r \in \mathbb{N}$  and every  $q > 1$ ,

$$\sum_{\xi \bmod q}^* \overline{\xi(a)} \sum_{n \leq x} \xi(n) f_r(n) = \sum_{d|q} \mu(q/d) \varphi(d) \Delta_{f_{rq}}(x; d, a).$$

*Proof. Step 1.* Since  $d \mid q$ , the reduction of  $a$  modulo  $d$  is a unit, so Lemma 3.22 gives  $\varphi(d) \Delta_g(x; d, a) = \sum_{\chi \neq \chi_0 \bmod d} \overline{\chi(a)} \sum_{n \leq x} \chi(n) g(n)$ . Apply Lemma 9.1 to the modulus  $d$ . For  $\xi$  primitive modulo  $i \mid d$  and  $n$  with  $g(n) \neq 0$  we have  $(n, q) = 1$ , hence  $(n, d) = 1$  and  $(\xi \uparrow^d)(n) = \xi(n)$ ; likewise  $(\xi \uparrow^d)(a) = \xi(a)$  because  $a$  is a unit. So the summand attached to  $\xi \uparrow^d$  is exactly the summand of  $G_i$ .

*Step 2.* Put  $g := f_{rq}$ ; then  $g(n) = 0$  whenever  $(n, q) > 1$ . Since  $\xi(n) = 0$  unless  $(n, q) = 1$ , and  $f_r(n) = f_{rq}(n)$  when  $(n, q) = 1$ , the left-hand side of the second identity equals  $G_q$ . Define  $G'_i := G_i$  for  $i \mid q$ ,  $i \neq 1$ , and  $G'_1 := 0$ , and set  $F(d) := \sum_{i|d} G'_i$  for  $d \mid q$ . By Step 1,  $F(d) = \varphi(d) \Delta_g(x; d, a)$ . Möbius inversion on the divisors of  $q$  (Lemma 3.9) gives  $G'_q = \sum_{d|q} \mu(q/d) F(d)$ , and  $G'_q = G_q$  because  $q \neq 1$ .  $\square$

**Definition 9.3** (The twisted sums  $S_r(y, \xi)$ ). For  $r \in \mathbb{N}$ , a real  $y$  and a Dirichlet character  $\xi$  modulo some  $d \in \mathbb{N}$ , set

$$S_r(y, \xi) := \left| \sum_{n \leq y} \Lambda_r^b(n) \xi(n) \right| \geq 0,$$

where  $\Lambda_r^b$  is the restriction of  $\Lambda^b$  to the integers coprime to  $r$ .

**Proposition 9.4** (Decomposing  $\Delta_{\Lambda^b}$  by conductor). For every  $C \in \mathbb{N}$ , every real  $y$ , every  $q \geq 1$  and every  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ ,

$$|\Delta_{\Lambda^b}(y; q, a)| \leq \frac{1}{\varphi(q)} \left| \sum_{\substack{d|q \\ 1 < d \leq (\log x)^C}} \sum_{s|d} \mu(d/s) \varphi(s) \Delta_{\Lambda_q^b}(y; s, a) \right| + \frac{1}{\varphi(q)} \sum_{\substack{d|q \\ d > (\log x)^C}} \sum_{\xi \bmod d}^* S_{q/d}(y, \xi).$$

*Proof.* By Lemma 3.20,  $\Delta_{\Lambda^b}(y; q, a) = \Delta_{\Lambda_q^b}(y; q, a)$ , and the first part of Lemma 9.2 with  $g = \Lambda_q^b$  and  $d = q$  gives

$$\varphi(q) \Delta_{\Lambda^b}(y; q, a) = \sum_{\substack{d|q \\ d \neq 1}} G_d, \quad G_d = \sum_{\xi \bmod d}^* \overline{\xi(a)} \sum_{n \leq y} \xi(n) \Lambda_q^b(n).$$

Split the divisors  $d \neq 1$  of  $q$  according to whether  $d \leq (\log x)^C$  or  $d > (\log x)^C$  and apply the triangle inequality to the two groups.

For  $1 < d \leq (\log x)^C$ , the second part of Lemma 9.2 (with modulus  $d$ ,  $r = q$  and  $f = \Lambda^b$ ) expresses  $G_d$  as  $\sum_{s|d} \mu(d/s) \varphi(s) \Delta_{\Lambda_{qd}^b}(y; s, a)$ , and  $\Lambda_{qd}^b = \Lambda_q^b$  since  $d \mid q$ . This is the first term.

For  $d > (\log x)^C$  we bound  $|G_d|$  termwise:  $|\xi(a)| = 1$ , and for a character  $\xi$  modulo  $d$  one has  $\xi(n) \Lambda_q^b(n) = \xi(n) \Lambda_{q/d}^b(n)$  for all  $n$ , because both sides vanish unless  $(n, d) = 1$ , in which case  $(n, q) = 1 \iff (n, q/d) = 1$ . Hence  $|G_d| \leq \sum_{\xi}^* S_{q/d}(y, \xi)$ , which is the second term.  $\square$

## 9.2 Small conductors

For conductors  $d \leq (\log x)^C$  the Siegel–Walfisz theorem applies to every modulus  $s \mid d$ . We bound the discrepancy of  $\Lambda_q^b$  by writing  $\Lambda_q^b = \Lambda_q - \Lambda_q^\sharp - (\Lambda_{\leq U})_q$  (Vaughan’s identity restricted to integers coprime to  $q$ ) and bounding the three pieces separately: the first by Siegel–Walfisz, the second by the pointwise Type I bound of Proposition 8.7, and the third trivially. Here it is essential that the Type I bound is available for the *restricted* function  $\Lambda_q^\sharp$ , and the factor  $\tau(q)$  it carries is absorbed using  $q \leq \sqrt{x}$ .

**Lemma 9.5** (Siegel–Walfisz for  $\Lambda_q$ ). *Let  $A', C_2 \in \mathbb{N}$ . For every real  $2 \leq y \leq x$ , all integers  $1 \leq s \mid q$  with  $q \leq \sqrt{x}$  and  $s \leq (\log y)^{C_2}$ , and every  $a \in (\mathbb{Z}/s\mathbb{Z})^*$ ,*

$$|\Delta_{\Lambda_q}(y; s, a)| \ll_{A', C_2} \frac{y}{(\log y)^{A'}} + (\log x)^2.$$

*Proof.* Write  $\Lambda_q = \Lambda - \Lambda^{\text{nc}}$  where  $\Lambda^{\text{nc}}(n) = \Lambda(n)1_{(n, q) > 1}$ , and use Lemma 3.18.

*Step 1: the non-coprime part.* Since  $\Lambda^{\text{nc}} \geq 0$ , Lemma 3.19 and Lemma 3.7 give  $|\Delta_{\Lambda^{\text{nc}}}(y; s, a)| \leq 2 \sum_{n \leq y, (n, q) > 1} \Lambda(n) \ll \log q \log y \leq (\log x)^2$ , using  $q \leq x$  and  $y \leq x$ .

*Step 2: the main part.* By Lemma 3.21,

$$\Delta_{\Lambda}(y; s, a) = \left( \psi(y; s, a) - \frac{y}{\varphi(s)} \right) + \frac{1}{\varphi(s)} \left( y - \sum_{\substack{n \leq y \\ (n, s) = 1}} \Lambda(n) \right).$$

The first bracket is  $\ll_{A', C_2} y/(\log y)^{A'}$  by Theorem 1, since  $1 \leq s \leq (\log y)^{C_2}$  and  $y \geq 2$ . The second is bounded, using  $1/\varphi(s) \leq 1$  and Corollary 4.1 with  $A = A'$ , by  $\ll_{A'} y/(\log y)^{A'} + \log s \log y$ , and  $\log s \log y \leq (\log x)^2$ . Adding the three contributions gives the claim.  $\square$

**Lemma 9.6** (Siegel–Walfisz for  $\Lambda_q^b$ ). *Let  $A, C \in \mathbb{N}$ . For every real  $\sqrt{x} \leq y \leq x$ , all integers  $1 \leq s \mid q$  with  $q \leq \sqrt{x}$  and  $s \leq (\log x)^C$ , and every  $a \in (\mathbb{Z}/s\mathbb{Z})^*$ ,*

$$|\Delta_{\Lambda_q^b}(y; s, a)| \ll_{A, C} \frac{x}{(\log x)^{A+2C+1}}.$$

*Proof.* Put  $A' := A + 2C + 1$ . By Vaughan’s identity (Proposition 6.2) and linearity of restriction (Lemma 3.13),  $\Lambda_q^b = \Lambda_q - \Lambda_q^\sharp - (\Lambda_{\leq U})_q$ , so by Lemma 3.18 it suffices to bound the three discrepancies separately by  $\ll_{A, C} x/(\log x)^{A'}$ . Note  $y \geq \sqrt{x} \geq 2$  and  $\log y \geq \frac{1}{2} \log x \geq 8$  (Lemma 3.24).

*Step 1:  $\Lambda_q$ .* Since  $(\log y)^2 \geq \frac{1}{4}(\log x)^2 \geq \log x$ , we have  $s \leq (\log x)^C \leq (\log y)^{2C}$ , so Lemma 9.5 applies with  $C_2 = 2C$ . By Lemma 5.4(2),  $y/(\log y)^{A'} \ll_{A'} x/(\log x)^{A'}$ , and by Lemma 5.5 (first bound with  $j = 2$ ,  $N = A'$ ),  $(\log x)^2 \ll_{A'} x/(\log x)^{A'}$ .

*Step 2:  $\Lambda_q^\sharp$ .* The reduction of  $a$  modulo  $s$  is a unit and  $q \leq \sqrt{x} \leq x$ , so Proposition 8.7 with  $r = q$  gives  $|\Delta_{\Lambda_q^\sharp}(y; s, a)| \ll \tau(q)UV \log x \leq \tau(q)\sqrt{x} \log x$ , and the third bound of Lemma 5.5 with  $N = A'$  gives  $\ll_{A'} x/(\log x)^{A'}$ .

*Step 3:  $(\Lambda_{\leq U})_q$ .* By Lemma 7.2,  $|\Delta_{(\Lambda_{\leq U})_q}(y; s, a)| \leq 2U \log x \leq 2\sqrt{x} \log x$ , and the second bound of Lemma 5.5 with  $j = 1$ ,  $N = A'$  gives  $\ll_{A'} x/(\log x)^{A'}$ .  $\square$

**Proposition 9.7** (The small conductors). *Let  $A, C \in \mathbb{N}$ . For every real  $\sqrt{x} \leq y \leq x$ , every*

integer  $1 \leq q \leq \sqrt{x}$  and every  $a \in (\mathbb{Z}/q\mathbb{Z})^*$ ,

$$\left| \sum_{\substack{d|q \\ 1 < d \leq (\log x)^C}} \sum_{s|d} \mu(d/s) \varphi(s) \Delta_{\Lambda_q^\flat}(y; s, a) \right| \ll_{A,C} \frac{x}{(\log x)^{A+1}}.$$

*Proof.* Push the absolute value inside and use  $|\mu(d/s)| \leq 1$ . For  $s \mid d \mid q$  the reduction of  $a$  modulo  $s$  is a unit and  $s \leq d \leq (\log x)^C$ , so Lemma 9.6 bounds each discrepancy by  $\ll_{A,C} x/(\log x)^{A+2C+1}$ . Since  $\sum_{s|d} \varphi(s) = d$  (Lemma 3.10), the left-hand side is

$$\ll_{A,C} \frac{x}{(\log x)^{A+2C+1}} \sum_{\substack{d|q \\ 1 < d \leq (\log x)^C}} d \leq \frac{x}{(\log x)^{A+2C+1}} \cdot (\log x)^C \cdot (\log x)^C,$$

because there are at most  $(\log x)^C$  such  $d$ , each at most  $(\log x)^C$ .  $\square$

### 9.3 Reduction to the quantities $T_r$

**Definition 9.8** (The quantities  $T_r(Q)$ ). For a real  $C$ , an integer  $r \geq 1$  and a real  $Q$ , set

$$T_r^{(C)}(Q) := \sum_{(\log x)^C < d \leq Q/r} \frac{1}{\varphi(d)} \sum_{\xi \bmod d}^* \max_{\sqrt{x} \leq y \leq x} S_r(y, \xi) \in [0, \infty],$$

the maximum being that of Definition 3.25. For  $r = 0$  the range of  $d$  is empty and  $T_r^{(C)}(Q) = 0$ . We usually suppress  $C$  from the notation.

**Proposition 9.9** (Reduction of the Type II estimate to  $T_r$ ). Let  $A, C \in \mathbb{N}$ . For every integer  $1 \leq q \leq \sqrt{x}$ ,

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^\flat}(y; q, a)| \leq \frac{1}{\varphi(q)} \sum_{\substack{d|q \\ d > (\log x)^C}} \sum_{\xi \bmod d}^* \max_{\sqrt{x} \leq y \leq x} S_{q/d}(y, \xi) + O_{A,C} \left( \frac{1}{\varphi(q)} \frac{x}{(\log x)^{A+1}} \right),$$

and consequently, for every real  $Q \leq \sqrt{x}$ ,

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^\flat}(y; q, a)| \leq \sum_{r \leq Q} \frac{T_r^{(C)}(Q)}{\varphi(r)} + O_{A,C} \left( \frac{x}{(\log x)^A} \right).$$

*Proof. Step 1: a fixed modulus.* By Lemma 3.26(2) it suffices to bound  $|\Delta_{\Lambda^\flat}(y; q, a)|$  for each unit  $a$  and each  $\sqrt{x} \leq y \leq x$ . Apply Proposition 9.4, bound the first term by Proposition 9.7, and bound each  $S_{q/d}(y, \xi)$  in the second term by its maximum over  $y$  (Lemma 3.26(1)).

*Step 2: the error term.* Sum the bound of Step 1 over  $1 \leq q \leq Q$  (each such  $q$  satisfies  $q \leq \sqrt{x}$ ). The error terms total  $\ll_{A,C} x(\log x)^{-A-1} \sum_{q \leq Q} 1/\varphi(q)$ , and Lemma 5.3 (applicable as  $Q \leq \sqrt{x} \leq x$ , Lemma 3.24) bounds the sum by  $\ll \log x$ , giving  $\ll_{A,C} x/(\log x)^A$ .

*Step 3: the main term.* For  $d \mid q$  write  $q = dr$ . Since  $\varphi(d)\varphi(r) \leq \varphi(dr)$  (Lemma 3.10),  $1/\varphi(q) \leq \varphi(d)^{-1}\varphi(r)^{-1}$ . Hence the main terms total at most

$$\sum_{q \leq Q} \sum_{\substack{d|q \\ d > (\log x)^C}} \frac{1}{\varphi(r)} \frac{1}{\varphi(d)} \sum_{\xi \bmod d}^* \max_y S_r(y, \xi), \quad r = q/d.$$

The map  $(q, d) \mapsto (r, d) = (q/d, d)$  is injective, and its image is contained in the set of pairs with  $1 \leq r \leq Q$  and  $(\log x)^C < d \leq Q/r$ . Since all terms are non-negative, the sum is at most the sum over that larger set, which is  $\sum_{r \leq Q} T_r^{(C)}(Q)/\varphi(r)$  by Definition 9.8.  $\square$

## 9.4 A maximal large sieve for convolutions

The remaining input is a maximal version of the large sieve inequality for Dirichlet convolutions (Proposition 9.21), Theorem 26.6 of [4]. The textbook proof uses the truncated Perron formula. We use instead a smoothed cutoff  $\tilde{1}_\varepsilon$ , whose Mellin transform is  $s^{-1}\mathcal{M}[\nu](\varepsilon s)$  for a fixed bump function  $\nu$ . The point is that for  $y = K + \frac{1}{2}$  a half-integer and  $\varepsilon = 1/(6 \log 2 \cdot x)$ , the smoothed cutoff coincides with the sharp cutoff at every integer (Lemma 9.20), so the smoothed bilinear sums  $T_\varepsilon(y, \chi)$  are *equal* to the partial sums we want to bound, and no error term needs to be estimated. Throughout this section  $x \geq 1$  and  $Q \geq 1$  are arbitrary reals,  $(f, g)$  is a supported pair and  $\nu$  a bump function; no parameter datum is involved. We write  $\mathcal{M}[F](s) := \int_0^\infty u^{s-1} F(u) du$  for the Mellin transform.

**Definition 9.10** (Supported pairs and Dirichlet polynomials). A *supported pair* consists of complex arithmetic functions  $f, g$  and real numbers  $M, N > 0$  with  $f(n) = 0$  for  $n > M$  and  $g(n) = 0$  for  $n > N$ . For such a pair, a character  $\chi$  and  $s \in \mathbb{C}$  we write

$$F_s(\chi) := \sum_{m \leq M} f(m) \chi(m) m^{-s}, \quad G_s(\chi) := \sum_{n \leq N} g(n) \chi(n) n^{-s}$$

for the associated Dirichlet polynomials, and  $\|f\|_2 := (\sum_{m \leq M} |f(m)|^2)^{1/2}$ ,  $\|g\|_2 := (\sum_{n \leq N} |g(n)|^2)^{1/2}$ .

**Proposition 9.11** (The smoothed cutoff). *Let  $\nu : \mathbb{R} \rightarrow \mathbb{R}$  be a  $C^1$  function supported in  $[1/2, 2]$ . For  $\varepsilon > 0$  define the smoothed cutoff  $\tilde{1}_\varepsilon : \mathbb{R} \rightarrow \mathbb{R}$  as the multiplicative convolution of the indicator of  $(0, 1]$  with the spike  $t \mapsto \nu(t^{1/\varepsilon})/\varepsilon$ :*

$$\tilde{1}_\varepsilon(u) := \int_0^\infty 1_{(0,1]}(t) \frac{1}{\varepsilon} \nu((u/t)^{1/\varepsilon}) \frac{dt}{t}.$$

1. *There exists a  $C^\infty$  function  $\nu$  supported in  $[1/2, 2]$  with  $\nu \geq 0$  and  $\int_0^\infty \nu(t) dt/t = 1$ .*
2. *If  $\int_0^\infty \nu(t) dt/t = 1$ , then  $\tilde{1}_\varepsilon(u) = 1$  for  $\varepsilon > 0$  and  $0 < u \leq 1 - (\log 2)\varepsilon$ .*
3. *If  $0 < \varepsilon < 1$ , then  $\tilde{1}_\varepsilon(u) = 0$  for  $u \geq 1 + 2(\log 2)\varepsilon$ .*
4. *If  $\nu \geq 0$  on  $(0, \infty)$ , then  $\tilde{1}_\varepsilon(u) \geq 0$  for all  $\varepsilon > 0$  and  $u > 0$ ; if moreover  $\int_0^\infty \nu(t) dt/t = 1$ , then  $\tilde{1}_\varepsilon(u) \leq 1$ .*
5. *If  $\nu \geq 0$  on  $(0, \infty)$ , then  $\tilde{1}_\varepsilon$  is continuous at every  $u > 0$ .*
6. *For  $\varepsilon > 0$  and  $\Re s > 0$ ,  $\mathcal{M}[\tilde{1}_\varepsilon](s) = s^{-1} \mathcal{M}[\nu](\varepsilon s)$ .*
7. *There is a constant  $C > 0$ , depending only on  $\nu$ , such that for all  $\sigma_1 > 0$ , all  $s$  with  $\sigma_1 \leq \Re s \leq 2$  and all  $0 < \varepsilon < 1$ ,  $|\mathcal{M}[\tilde{1}_\varepsilon](s)| \leq C/(\varepsilon|s|^2)$ .*

*Proof.* The definition and all seven properties are taken from the PrimeNumberTheoremAnd library [6] (module `MellinCalculus`), where they are proved; the proofs are not reproduced here.  $\square$

**Definition 9.12** (Bump functions). A *bump function* is a function  $\nu : \mathbb{R} \rightarrow \mathbb{R}$  that is  $C^\infty$ , supported in  $[1/2, 2]$ , non-negative, and normalised by  $\int_0^\infty \nu(t) \frac{dt}{t} = 1$ . Such functions exist by Proposition 9.11(1); one is fixed once and for all, and the implied constants in this section may depend on this choice. For  $\varepsilon > 0$  we write  $\tilde{1}_\varepsilon$  for the smoothed cutoff of Proposition 9.11 built from  $\nu$ ; by that proposition,  $0 \leq \tilde{1}_\varepsilon \leq 1$ ,  $\tilde{1}_\varepsilon = 1$  on  $(0, 1 - (\log 2)\varepsilon]$ ,  $\tilde{1}_\varepsilon = 0$  on  $[1 + 2(\log 2)\varepsilon, \infty)$  when  $\varepsilon < 1$ , and  $\mathcal{M}[\tilde{1}_\varepsilon](s) = s^{-1}\mathcal{M}[\nu](\varepsilon s)$  for  $\Re s > 0$ .

**Lemma 9.13** (The Mellin transform of a bump is bounded on strips). *Let  $\nu \in C^1(\mathbb{R})$  be supported in  $[1/2, 2]$  and let  $\sigma_1 \leq \sigma_2$  with  $\sigma_2 \geq 0$ . Then there is a constant  $C$  such that  $|\mathcal{M}[\nu](s)| \leq C|s|^{-1}$  for all  $s \neq 0$  with  $\sigma_1 \leq \Re s \leq \sigma_2$ ; consequently  $\mathcal{M}[\nu]$  is bounded on the whole strip  $\{\sigma_1 \leq \Re s \leq \sigma_2\}$ .*

*Proof.* Integration by parts (valid since  $\nu$  has compact support in  $(0, \infty)$ ) gives  $\mathcal{M}[\nu](s) = \int_0^\infty \nu(u)u^{s-1}du = -\frac{1}{s} \int_0^\infty \nu'(u)u^s du$  for  $s \neq 0$ . On  $[1/2, 2]$  we have  $|u^s| = u^{\Re s} \leq \max(2^{\sigma_2}, 2^{-\sigma_1})$ , and  $|\nu'|$  is bounded by its maximum on  $[1/2, 2]$ ; the interval has length  $3/2$ . This gives the first bound with  $C = \frac{3}{2} \max|\nu'| \cdot \max(2^{\sigma_2}, 2^{-\sigma_1})$ . For boundedness on the strip, note that  $\mathcal{M}[\nu]$  is entire (the Mellin integral of a compactly supported  $C^1$  function converges for every  $s$ ), hence bounded near  $s = 0$ ; away from a small ball around 0 the first bound applies.  $\square$

**Lemma 9.14** (The Mellin transform of the smoothed cutoff). *Let  $\nu$  be a bump function,  $0 < \varepsilon < 1$  and  $0 < \sigma \leq 2$ .*

1. *For all  $t \in \mathbb{R}$ , with  $s = \sigma + it$ ,*

$$|\mathcal{M}[\tilde{1}_\varepsilon](s)| \ll \frac{1}{|s|} \quad \text{and} \quad |\mathcal{M}[\tilde{1}_\varepsilon](s)| \ll \frac{1}{\varepsilon|s|^2},$$

*the implied constants depending only on  $\nu$ .*

2. *The function  $t \mapsto \mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)$  is integrable on  $\mathbb{R}$ , and for every  $u > 0$ ,*

$$\tilde{1}_\varepsilon(u) = \frac{1}{2\pi} \int_{-\infty}^\infty u^{-(\sigma+it)} \mathcal{M}[\tilde{1}_\varepsilon](\sigma + it) dt.$$

*Proof.* (1) For the first bound use  $\mathcal{M}[\tilde{1}_\varepsilon](s) = s^{-1}\mathcal{M}[\nu](\varepsilon s)$  (Proposition 9.11(6)) and the boundedness of  $\mathcal{M}[\nu]$  on the strip  $0 \leq \Re s \leq 2$  (Lemma 9.13); note  $\Re(\varepsilon s) = \varepsilon\sigma \in (0, 2)$ . The second bound is Proposition 9.11(7) with lower edge  $\sigma_1 = \sigma$ . Only these two bounds are needed; in particular no rapid decay of  $\mathcal{M}[\nu]$  is used.

(2) By the second bound in (1) the integrand is dominated by a constant multiple of  $(\sigma^2 + t^2)^{-1}$ , which is integrable; measurability follows from continuity of  $\mathcal{M}[\tilde{1}_\varepsilon]$  on  $\Re s > 0$ . The inversion formula is then the Mellin inversion theorem, whose hypotheses are: convergence of the Mellin integral of  $\tilde{1}_\varepsilon$  at  $\sigma$ , integrability of the transform along the vertical line (just shown), and continuity of  $\tilde{1}_\varepsilon$  at  $u$  (Proposition 9.11(5)).  $\square$

**Definition 9.15** (The smoothed bilinear sums  $T_\varepsilon(y, \chi)$ ). For a supported pair  $(f, g)$ , a bump function  $\nu$ ,  $\varepsilon > 0$ ,  $y > 0$  and a Dirichlet character  $\chi$ , set

$$T_\varepsilon(y, \chi) := \sum_{m \leq M} \sum_{n \leq N} f(m)\chi(m) g(n)\chi(n) \tilde{1}_\varepsilon\left(\frac{mn}{y}\right).$$

**Lemma 9.16** (Mellin representation of  $T_\varepsilon$ ). *Let  $(f, g)$  be a supported pair,  $\nu$  a bump function,  $0 < \varepsilon < 1$ ,  $0 < \sigma \leq 2$ ,  $y \geq 1$  and  $\chi$  a Dirichlet character. Then*

$$T_\varepsilon(y, \chi) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F_{\sigma+it}(\chi) G_{\sigma+it}(\chi) y^{\sigma+it} \mathcal{M}[\tilde{1}_\varepsilon](\sigma+it) dt,$$

and consequently

$$|T_\varepsilon(y, \chi)| \leq \frac{y^\sigma}{2\pi} \int_{-\infty}^{\infty} |F_{\sigma+it}(\chi)| |G_{\sigma+it}(\chi)| |\mathcal{M}[\tilde{1}_\varepsilon](\sigma+it)| dt,$$

where the last integrand is integrable.

*Proof.* Apply the inversion formula of Lemma 9.14(2) to each term of  $T_\varepsilon(y, \chi)$  at  $u = mn/y > 0$ :

$$T_\varepsilon(y, \chi) = \sum_{m \leq M} \sum_{n \leq N} f(m) \chi(m) g(n) \chi(n) \cdot \frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{mn}{y}\right)^{-(\sigma+it)} \mathcal{M}[\tilde{1}_\varepsilon](\sigma+it) dt.$$

Each integrand is a continuous function of constant modulus  $|f(m)|m^{-\sigma}|g(n)|n^{-\sigma}y^\sigma$  times the integrable function  $\mathcal{M}[\tilde{1}_\varepsilon](\sigma+it)$ , hence integrable, so the finite double sum can be moved inside the integral. Using  $(mn/y)^{-s} = m^{-s}n^{-s}y^s$  for positive reals and collecting the sums gives  $F_s(\chi)G_s(\chi)y^s$ . The inequality follows by taking absolute values inside the integral and using  $|y^{\sigma+it}| = y^\sigma$ ; integrability of the majorant follows since  $|F_{\sigma+it}| |G_{\sigma+it}|$  is continuous and bounded by  $\sum_{m \leq M} |f(m)|m^{-\sigma} \cdot \sum_{n \leq N} |g(n)|n^{-\sigma}$ .  $\square$

**Lemma 9.17** (The large sieve for Dirichlet polynomials). *Let  $Q \geq 1$ ,  $\sigma > 0$  and  $t \in \mathbb{R}$ .*

1. *For every  $h : \mathbb{N} \rightarrow \mathbb{C}$  and every  $P \in \mathbb{R}$ ,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \left| \sum_{m \leq P} h(m) \chi(m) m^{-(\sigma+it)} \right|^2 \ll (P + Q^2) \sum_{m \leq P} |h(m)|^2.$$

2. *For every supported pair  $(f, g)$ ,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} |F_{\sigma+it}(\chi)| |G_{\sigma+it}(\chi)| \ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) \|f\|_2 \|g\|_2.$$

The implied constants are those of Theorem 2; in particular they are uniform in  $t$  and  $\sigma$ .

*Proof.* (1) If  $P < 1$  the left side is 0. Otherwise apply Theorem 2 with  $H = 0$ ,  $N = \lfloor P \rfloor$  and  $c_n = h(n)n^{-(\sigma+it)}$  for  $1 \leq n \leq N$ . Since  $|c_n|^2 = |h(n)|^2 n^{-2\sigma} \leq |h(n)|^2$  for  $n \geq 1$ , and  $\lfloor P \rfloor \leq P$ , the right-hand side of the large sieve inequality is  $\ll (P + Q^2) \sum_{m \leq P} |h(m)|^2$ .

(2) By the Cauchy–Schwarz inequality with weights  $w_{q,\chi} = q/\varphi(q) \geq 0$  over the pairs  $(q, \chi)$ ,

$$\sum_{q,\chi} w_{q,\chi} |F| |G| \leq \left( \sum_{q,\chi} w_{q,\chi} |F|^2 \right)^{1/2} \left( \sum_{q,\chi} w_{q,\chi} |G|^2 \right)^{1/2}.$$

By (1) (with  $h = f$ ,  $P = M$ , and with  $h = g$ ,  $P = N$ ) the two factors are  $\ll \sqrt{M + Q^2} \|f\|_2$  and  $\ll \sqrt{N + Q^2} \|g\|_2$ . Finally  $\sqrt{M + Q^2} \sqrt{N + Q^2} \leq (\sqrt{M} + Q)(\sqrt{N} + Q) = \sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2$ , using  $\sqrt{a + Q^2} \leq \sqrt{a} + Q$  for  $a, Q \geq 0$ .  $\square$

**Lemma 9.18** (The kernel integral). *Let  $\nu$  be a bump function. Let  $x \geq 1$ ,  $\sigma := 1/\log(x+1)$  and  $\varepsilon := \frac{1}{6\log 2} \cdot \frac{1}{x}$  (so that  $0 < \varepsilon < 1$  and  $0 < \sigma \leq 2$ ). Then*

$$J := \int_{-\infty}^{\infty} |\mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)| dt \ll \log(x+1),$$

the implied constant depending only on  $\nu$ .

*Proof.* Write  $L = \log(x+1) \geq \log 2$  and split the integral at  $|t| = T := 1/\varepsilon$ .

*Step 1: the centre  $|t| \leq T$ .* By the first bound of Lemma 9.14(1) and the elementary inequality  $|\sigma + it|^{-1} \leq \sqrt{2}(\sigma + |t|)^{-1}$  (equivalent to  $(\sigma - |t|)^2 \geq 0$ ), the integrand is  $\ll (\sigma + |t|)^{-1}$ , and  $\int_{-T}^T (\sigma + |t|)^{-1} dt = 2 \log((\sigma + T)/\sigma) = 2 \log(1 + 6 \log 2 \cdot xL)$ . Since  $1 + 6 \log 2 \cdot xL \leq (1+x)(1 + 6 \log 2 \cdot L)$  and  $\log(1+u) \leq u$ , this is  $\ll L$ .

*Step 2: the tails  $|t| > T$ .* By the second bound of Lemma 9.14(1) and  $t^2 \leq |\sigma + it|^2$ , the integrand is  $\ll \varepsilon^{-1} t^{-2}$ , and  $\int_{|t|>T} t^{-2} dt = 2/T = 2\varepsilon$ . The tails contribute  $\ll 1 \ll L$ .  $\square$

**Proposition 9.19** (The smoothed bilinear sums on average). *Let  $(f, g)$  be a supported pair,  $\nu$  a bump function,  $Q \geq 1$ ,  $x \geq 1$  and  $\varepsilon := \frac{1}{6\log 2} \cdot \frac{1}{x}$ . Then*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x+1} |T_\varepsilon(y, \chi)| \ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) \|f\|_2 \|g\|_2 \log(x+1),$$

and the same bound holds with the supremum replaced by  $\max_{1 \leq K \leq [x]} |T_\varepsilon(K + \frac{1}{2}, \chi)|$ . The implied constant depends only on  $\nu$  and on the constant in Theorem 2.

*Proof.* Put  $L := \log(x+1) \geq \log 2$  and  $\sigma := 1/L$ , so that  $0 < \sigma \leq 2$  and  $(x+1)^\sigma = e$ ; hence  $y^\sigma \leq e$  for  $1 \leq y \leq x+1$ . Also  $\varepsilon \leq 1/(6 \log 2) < 1$ . For each pair  $(q, \chi)$ , Lemma 9.16 gives the  $y$ -free majorant

$$\sup_{1 \leq y \leq x+1} |T_\varepsilon(y, \chi)| \leq \frac{e}{2\pi} \int_{\mathbb{R}} |F_{\sigma+it}(\chi)| |G_{\sigma+it}(\chi)| |\mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)| dt.$$

Multiply by  $q/\varphi(q)$ , sum over the finitely many  $(q, \chi)$ , and interchange the sum with the integral (all integrands are integrable). The resulting integrand is  $D(t) |\mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)|$  with  $D(t) = \sum_{q, \chi} \frac{q}{\varphi(q)} |F_{\sigma+it}(\chi)| |G_{\sigma+it}(\chi)|$ , which by Lemma 9.17(2) is  $\ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) \|f\|_2 \|g\|_2$  uniformly in  $t$ . Pulling this bound out, the remaining integral is  $J \ll L$  by Lemma 9.18.

For the discrete version note that  $K + \frac{1}{2} \in [1, x+1]$  for  $1 \leq K \leq [x]$ , so the maximum over  $K$  is at most the supremum over  $y$ .  $\square$

**Lemma 9.20** (The smoothed sum is sharp at half-integers). *Let  $x \geq 1$ , let  $(f, g)$  be a supported pair and let  $\chi$  be a Dirichlet character.*

1. *For every  $h : \mathbb{N} \rightarrow \mathbb{C}$ ,*

$$\sup_{1 \leq y \leq x} \left| \sum_{n \leq y} h(n) \right| = \max_{1 \leq K \leq [x]} \left| \sum_{n \leq K+1/2} h(n) \right|.$$

2. *For every real  $y$ ,*

$$\sum_{n \leq y} (f * g)(n) \chi(n) = \sum_{m \leq M} \sum_{n \leq N} 1_{mn \leq y} f(m) \chi(m) g(n) \chi(n).$$

3. Let  $\nu$  be a bump function and  $0 < \varepsilon \leq \frac{1}{6 \log 2} \cdot \frac{1}{x}$ . Then for every integer  $0 \leq K \leq x$ ,

$$T_\varepsilon(K + \tfrac{1}{2}, \chi) = \sum_{m \leq M} \sum_{n \leq N} 1_{mn \leq K+1/2} f(m) \chi(m) g(n) \chi(n) = \sum_{n \leq K+1/2} (f * g)(n) \chi(n).$$

In words: the supremum over  $y \in [1, x]$  of the partial sums of  $(f * g)\chi$  is attained at a half-integer, and at half-integers the smoothed cutoff  $\tilde{1}_\varepsilon(mn/(K + \frac{1}{2}))$  agrees with the sharp cutoff  $1_{mn \leq K+1/2}$  at every integer  $mn$ , so that  $T_\varepsilon(K + \frac{1}{2}, \chi)$  is exactly the partial sum.

*Proof.* (1) The partial sum  $\sum_{n \leq y} h(n)$  depends only on  $\lfloor y \rfloor$ , and  $\lfloor K + 1/2 \rfloor = K$ . As  $y$  ranges over  $[1, x]$ ,  $\lfloor y \rfloor$  ranges over  $\{1, \dots, \lfloor x \rfloor\}$ , so both sides are the maximum of the same finite set of numbers.

(2) The hyperbola identity (Lemma 3.8) with  $F = f\chi$ ,  $G = g\chi$  (using complete multiplicativity of  $\chi$ ) writes the left side as  $\sum_{m \leq y} \sum_{n \leq y/m} f(m)\chi(m)g(n)\chi(n)$ . Enlarging both ranges to  $m, n \leq \lfloor y \rfloor + \lfloor M \rfloor + \lfloor N \rfloor$  and inserting the indicator  $1_{mn \leq y}$  changes nothing, and then the support conditions on  $f$  and  $g$  allow one to shrink the ranges to  $m \leq M$ ,  $n \leq N$ .

(3) The second equality is (2). For the first, compare the two double sums termwise; put  $u = mn/(K + \frac{1}{2})$  with  $mn \geq 1$  an integer. If  $mn \leq K + \frac{1}{2}$  then  $mn \leq K$ , so  $u \leq K/(K + \frac{1}{2}) = 1 - \frac{1}{2K+1}$ . Since  $(\log 2)\varepsilon \leq \frac{1}{6x} \leq \frac{1}{2K+1}$  (as  $2K + 1 \leq 6x$  for  $K \leq x$ ,  $x \geq 1$ ), we get  $u \leq 1 - (\log 2)\varepsilon$  and  $\tilde{1}_\varepsilon(u) = 1$  (Proposition 9.11(2)). If  $mn > K + \frac{1}{2}$  then  $mn \geq K + 1$ , so  $u \geq (K + 1)/(K + \frac{1}{2}) = 1 + \frac{1}{2K+1}$ . Since  $2(\log 2)\varepsilon \leq \frac{1}{3x} \leq \frac{1}{2K+1}$  (as  $4K + 2 \leq 6x$ ), we get  $u \geq 1 + 2(\log 2)\varepsilon$ , and also  $\varepsilon < 1$ , so  $\tilde{1}_\varepsilon(u) = 0$  (Proposition 9.11(3)).  $\square$

**Proposition 9.21** (Maximal large sieve for convolutions). *Let  $(f, g)$  be a supported pair and  $\nu$  a bump function. For all real  $x \geq 1$  and  $Q \geq 1$ ,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x} \left| \sum_{n \leq y} (f * g)(n) \chi(n) \right| \ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) \|f\|_2 \|g\|_2 \log(x + 1).$$

In particular, let  $M, N \in \mathbb{N}$  and let  $f, g$  be real arithmetic functions with  $f(n) = 0$  for  $n > M$  and  $g(n) = 0$  for  $n > N$ . Then for all real  $x \geq 2$  and  $Q \geq 1$ ,

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x} \left| \sum_{n \leq y} (f * g)(n) \chi(n) \right| \ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) (\log x) \left( \sum_{n \leq M} f(n)^2 \right)^{1/2} \left( \sum_{n \leq N} g(n)^2 \right)^{1/2}.$$

The implied constants depend only on the bump function and on the constant in Theorem 2.

*Proof.* This is Theorem 26.6 of [4], proved with a smoothed cutoff instead of the truncated Perron formula. By Lemma 9.20(1) the supremum over  $y \in [1, x]$  may be replaced by the maximum over integers  $1 \leq K \leq \lfloor x \rfloor$  of the partial sum at  $y = K + \frac{1}{2}$ , and by Lemma 9.20(3) with  $\varepsilon = \frac{1}{6 \log 2} \cdot \frac{1}{x}$  that partial sum equals  $T_\varepsilon(K + \frac{1}{2}, \chi)$  (note  $K \leq \lfloor x \rfloor \leq x$ ). Now apply the discrete form of Proposition 9.19.

For the second statement: if  $M = 0$  or  $N = 0$  then  $f * g = 0$  and the left side vanishes. Otherwise  $(f, g)$ , viewed as complex arithmetic functions, form a supported pair with the real parameters  $M, N > 0$ , and the first statement (for the fixed bump function of Definition 9.12) gives the bound with  $\log(x + 1)$  in place of  $\log x$ ; finally  $\log(x + 1) \leq \log(x^2) = 2 \log x$  for  $x \geq 2$ .  $\square$

## 9.5 Dyadic decomposition and the bound for $T_r$

We return to a fixed parameter datum. To apply Proposition 9.21 to  $\Lambda_r^b$  we must write it as a sum of convolutions  $f_j * g_j$  with  $f_j$  supported on a dyadic interval  $(2^{j-1}, 2^j]$  and  $g_j$  supported on an initial segment. The savings  $1/\sqrt{U}$  and  $1/\sqrt{V}$  come from the support conditions  $U < 2^j$  and  $2^j \leq 2x/V$  that Vaughan's identity imposes.

**Lemma 9.22** (Dyadic decomposition of  $\Lambda^b$ ). *For every integer  $n \leq x$ ,*

$$\Lambda^b(n) = \sum_{\substack{j \in \mathbb{N} \\ U < 2^j \leq 2x/V}} (f_j * g_j)(n), \quad f_j := (\Lambda_{>U} * 1) 1_{(2^{j-1}, 2^j]}, \quad g_j := \mu \cdot 1_{(V, x/2^{j-1}]}.$$

*Proof.* Expand  $\Lambda^b(n) = \sum_{de=n} (\Lambda_{>U} * 1)(d) \mu_{>V}(e)$  (Definition 6.1) and, on the right,  $\sum_j (f_j * g_j)(n) = \sum_{de=n} \sum_j f_j(d) g_j(e)$ . Fix a pair  $de = n$ . If  $d \leq U$  then  $(\Lambda_{>U} * 1)(d) = \sum_{k|d} \Lambda_{>U}(k) = 0$  and every  $f_j(d) = 0$ ; if  $e \leq V$  then  $\mu_{>V}(e) = 0$  and every  $g_j(e) = 0$ . Otherwise  $d > U \geq 1$  and  $e > V$ , and there is a unique  $j_0 \geq 1$  with  $2^{j_0-1} < d \leq 2^{j_0}$ . It lies in the summation range, since  $U < d \leq 2^{j_0}$  and  $2^{j_0} < 2d \leq 2n/e \leq 2x/V$ ; moreover  $e = n/d \leq x/d < x/2^{j_0-1}$ , so  $f_{j_0}(d) g_{j_0}(e) = (\Lambda_{>U} * 1)(d) \mu(e) = (\Lambda_{>U} * 1)(d) \mu_{>V}(e)$ , while  $f_j(d) = 0$  for  $j \neq j_0$ .  $\square$

**Proposition 9.23** (The large sieve bound for  $S_r$ ). *For every  $r \in \mathbb{N}$  and every real  $Q \geq 2$ ,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \max_{\sqrt{x} \leq y \leq x} S_r(y, \chi) \ll \left( x + \frac{Qx}{\sqrt{U}} + \frac{Qx}{\sqrt{V}} + Q^2 \sqrt{x} \right) (\log x)^3.$$

*Proof. Step 1: dyadic decomposition.* Restricting Lemma 9.22 to integers coprime to  $r$  and using that restriction is a convolution homomorphism (Lemma 3.13),  $\Lambda_r^b(n) = \sum_j ((f_j)_r * (g_j)_r)(n)$  for  $n \leq x$ , the sum being over the set  $\mathcal{J} = \{j : U < 2^j \leq 2x/V\}$ . Hence, for  $\sqrt{x} \leq y \leq x$  (so  $1 \leq y \leq x$ ),

$$S_r(y, \chi) \leq \sum_{j \in \mathcal{J}} \sup_{1 \leq z \leq x} \left| \sum_{n \leq z} ((f_j)_r * (g_j)_r)(n) \chi(n) \right| =: \sum_{j \in \mathcal{J}} \Gamma_{r,j}(\chi),$$

and by Lemma 3.26(2) the same bound holds for the maximum over  $y$ .

*Step 2: one dyadic block.* Fix  $j \in \mathcal{J}$  and put  $M_j := 2^j$ ,  $N_j := \lfloor x/2^{j-1} \rfloor = \lfloor 2x/M_j \rfloor$ . The function  $(f_j)_r$  is supported on  $[1, M_j]$  and, since  $0 \leq (\Lambda_{>U} * 1)(n) \leq \sum_{k|n} \Lambda(k) = \log n \leq 2 \log x$  (Lemma 3.9) for  $n \leq 2^j \leq x^2$ , it satisfies  $\|(f_j)_r\|_2^2 \leq M_j (2 \log x)^2$ . The function  $(g_j)_r$  is supported on  $[1, N_j]$  with  $|(g_j)_r| \leq 1$ , so  $\|(g_j)_r\|_2^2 \leq N_j \leq 2x/M_j$ . Thus  $\|(f_j)_r\|_2 \|(g_j)_r\|_2 \ll \log x \sqrt{x}$ . By Proposition 9.21 (with  $x \geq 2$ ,  $Q \geq 1$ ),

$$\sum_{q \leq Q} \sum_{\chi}^* \frac{q}{\varphi(q)} \Gamma_{r,j}(\chi) \ll (\sqrt{N_j M_j} + \sqrt{M_j} Q + \sqrt{N_j} Q + Q^2) (\log x)^2 \sqrt{x}.$$

Now  $\sqrt{N_j M_j} \leq \sqrt{2x}$ ;  $\sqrt{M_j} \sqrt{x} \ll x/\sqrt{V}$  because  $M_j \leq 2x/V$ ; and  $\sqrt{N_j} \sqrt{x} \ll x/\sqrt{M_j} \leq x/\sqrt{U}$  because  $M_j > U$ . Hence each block contributes  $\ll (\log x)^2 (x + Qx/\sqrt{U} + Qx/\sqrt{V} + Q^2 \sqrt{x})$ , uniformly in  $j$ .

*Step 3: the number of blocks.*  $\#\mathcal{J} \leq \log_2(2x) \ll \log x$ , since  $\log x \geq 1$  (Lemma 3.24).

Multiplying gives the claim. Note that the savings  $1/\sqrt{U}$  and  $1/\sqrt{V}$  are obtained termwise from  $U < 2^j \leq 2x/V$  rather than by summing a geometric series; this costs one factor  $\log x$  (the number of blocks).  $\square$

**Proposition 9.24** (The bound for  $T_r$ ). *For every integer  $C \geq 3$ , every  $r \in \mathbb{N}$  and every real  $2 \leq Q \leq x$ ,*

$$T_r^{(C)}(Q) \ll \frac{x}{(\log x)^{C-3}} + \frac{x(\log x)^4}{\sqrt{U}} + \frac{x(\log x)^4}{\sqrt{V}} + \frac{Q\sqrt{x}(\log x)^3}{r},$$

with an implied constant independent of  $C$  and  $r$ .

*Proof.* Write  $B(d) := \sum_{\xi \bmod d}^* \frac{d}{\varphi(d)} \max_y S_r(y, \xi) \geq 0$ , so that  $T_r(Q) = \sum_{(\log x)^C < d \leq Q/r} B(d)/d$ . If the range is empty (in particular if  $r = 0$ ) there is nothing to prove; otherwise  $r \geq 1$  and  $Q/r \leq x$ .

*Step 1: dyadic blocks in  $d$ .* Group the  $d$  in the range by  $j = \lfloor \log_2 d \rfloor$ , which ranges over  $[j_L, j_U]$  with  $j_L = \lfloor \log_2 \lfloor (\log x)^C \rfloor \rfloor$  and  $j_U = \lfloor \log_2 \lfloor Q/r \rfloor \rfloor$ ; each  $d$  in the block  $j$  satisfies  $2^j \leq d < 2^{j+1}$ , so  $1/d \leq 2^{-j}$ . Hence

$$T_r(Q) \leq \sum_{j=j_L}^{j_U} 2^{-j} \sum_{d \leq 2^{j+1}} B(d) \ll \sum_{j=j_L}^{j_U} 2^{-j} \left( x + \frac{2^{j+1}x}{\sqrt{U}} + \frac{2^{j+1}x}{\sqrt{V}} + 4^{j+1}\sqrt{x} \right) (\log x)^3$$

by Proposition 9.23 applied with  $Q' = 2^{j+1} \geq 2$ .

*Step 2: summing over  $j$ .* Since  $(\log x)^C < 2^{j_L+1}$ , we have  $\sum_{j \geq j_L} 2^{-j} \leq 2 \cdot 2^{-j_L} \leq 4/(\log x)^C$ , so the first term contributes  $\ll x(\log x)^{3-C}$ . The two middle terms are independent of  $j$ , and the number of  $j$  is at most  $j_U + 1 \leq \log_2(2Q/r) \ll \log x$  (as  $2^{j_U} \leq Q/r \leq x$  and  $\log x \geq 1$ ), so they contribute  $\ll x(\log x)^4/\sqrt{U}$  and the same with  $V$ . Finally  $\sum_{j \leq j_U} 2^{j+1} \leq 4 \cdot 2^{j_U} \leq 4Q/r$ , so the last term contributes  $\ll Q\sqrt{x}(\log x)^3/r$ .  $\square$

## 9.6 The Type II estimate

**Proposition 9.25** (The Type II part on average). *For every  $A \in \mathbb{N}$  and every real  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^*}(y; q, a)| \ll_A \frac{x}{(\log x)^A}.$$

*Proof.* Since  $\log x \geq 1$ ,  $Q \leq \sqrt{x} \leq x$ . Apply Proposition 9.9 with  $C = A + 4$ ; the error term is  $\ll_A x/(\log x)^A$ , and it remains to bound  $\sum_{r \leq Q} T_r^{(A+4)}(Q)/\varphi(r)$ .

*Step 1: small  $Q$ .* If  $Q < 2$  then for every  $r \geq 1$  we have  $Q/r < 2 \leq 16 \leq (\log x)^{A+4}$  (Lemma 3.24), so the range of  $d$  in  $T_r$  is empty and the main term vanishes.

*Step 2:  $Q \geq 2$ .* By Proposition 9.24 with  $C = A + 4$  (so  $C - 3 = A + 1$  and  $Q \leq x$ ),

$$\sum_{r \leq Q} \frac{T_r(Q)}{\varphi(r)} \ll \left( \frac{x}{(\log x)^{A+1}} + \frac{x(\log x)^4}{\sqrt{U}} + \frac{x(\log x)^4}{\sqrt{V}} \right) \sum_{r \leq Q} \frac{1}{\varphi(r)} + Q\sqrt{x}(\log x)^3 \sum_{r \leq Q} \frac{1}{r\varphi(r)}.$$

By Lemma 5.3,  $\sum_{r \leq Q} 1/\varphi(r) \ll \log x$ . The first product is then  $\ll x/(\log x)^A$ . For the second and third, Lemma 5.4(3) with  $n = A + 5$  and  $W = U$  (resp.  $V$ ) gives  $(\log x)^{A+5} \ll_A \sqrt{U}$ , i.e.  $x(\log x)^5/\sqrt{U} \ll_A x/(\log x)^A$ . For the last term, Lemma 5.2 bounds the sum by  $\ll 1$ , and  $Q\sqrt{x}(\log x)^3 \leq \sqrt{x}(\log x)^{-A-3}\sqrt{x}(\log x)^3 = x/(\log x)^A$ .  $\square$

## Chapter 10

# Proof of the main theorem

We now combine the three estimates for the pieces of Vaughan's identity into a bound for  $\Delta_\Lambda$  on average (Proposition 10.1), and then pass from  $\Delta_\Lambda$  back to the Chebyshev error  $\psi(y; q, a) - y/\varphi(q)$  using the prime number theorem. Two small matters remain: the parameter datum forces  $x \geq e^{16}$ , so the range  $x \leq e^{32}$  in which no admissible choice of  $U = V$  exists is handled by the trivial bound; and the maximum over  $y$  in the main theorem runs over all of  $[1, x]$ , while the estimates so far only cover  $\sqrt{x} \leq y \leq x$ .

**Proposition 10.1** (Bombieri–Vinogradov for  $\Delta_\Lambda$ ). *For every  $A \in \mathbb{N}$  and every real  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_\Lambda(y; q, a)| \ll_A \frac{x}{(\log x)^A}.$$

*Proof.* Sum Lemma 6.3 over  $q \leq Q$  and bound the three resulting sums by Proposition 8.9, Proposition 9.25 and Proposition 7.3; the last gives  $\ll x/(\log x)^{A+2} \leq x/(\log x)^A$ , as  $\log x \geq 1$  (Lemma 3.24).  $\square$

**Definition 10.2** (The maximal Chebyshev error). For a real  $x$  and  $q \in \mathbb{N}$  set

$$E(x; q) := \sup_{1 \leq y \leq x} \sup_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \in [0, \infty].$$

Here  $y$  ranges over all of  $[1, x]$ , not only over  $[\sqrt{x}, x]$ , and  $x$  is an arbitrary real (no parameter datum is involved). For  $q \geq 1$ ,  $E(x; q) \ll x$  by Lemma 3.6, so  $E(x; q)$  is finite and equals the supremum of the same quantities taken in  $\mathbb{R}$ .

**Lemma 10.3** (The compact range  $x \leq e^{32}$ ). *Let  $A \in \mathbb{N}$  and let  $2 \leq x \leq e^{32}$  and  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$  be real. Then*

$$\sum_{q \leq Q} E(x; q) \ll_A \frac{x}{(\log x)^A}.$$

*Proof.* By Lemma 3.6,  $E(x; q) \ll x$  for every  $q \geq 1$ , so the sum is  $\ll Qx$ . Since  $\sqrt{x} \leq e^{16}$  and  $\log x \geq \log 2$ , the hypothesis gives  $Q \leq e^{16}/(\log 2)^{A+3} \ll_A 1$ . Finally  $(\log x)^A \leq 32^A$ , i.e.  $1 \ll_A (\log x)^{-A}$ .  $\square$

**Lemma 10.4** (From  $\Delta_\Lambda$  to the Chebyshev error). *Let  $A \in \mathbb{N}$ . For every parameter datum and every real  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,*

$$\sum_{q \leq Q} E(x; q) \ll_A \frac{x}{(\log x)^A}.$$

*Proof.* Write  $L := \log x \geq 1$  and note  $Q \leq \sqrt{x} \leq x$  and  $\sqrt{x} \geq 2$  (Lemma 3.24).

*Step 1: the pointwise bound.* Fix  $1 \leq q \leq Q$ , a unit  $a$  and  $1 \leq z \leq x$ . If  $z < \sqrt{x}$ , then Lemma 3.6 gives  $|\psi(z; q, a) - z/\varphi(q)| \ll z \leq \sqrt{x}$ . If  $z \geq \sqrt{x}$  (so  $z \geq 2$ ), Lemma 3.21 gives

$$\psi(z; q, a) - \frac{z}{\varphi(q)} = \Delta_\Lambda(z; q, a) + \frac{1}{\varphi(q)} \left( \sum_{\substack{n \leq z \\ (n, q)=1}} \Lambda(n) - z \right),$$

and by Corollary 4.1 with  $A + 2$  in place of  $A$ , Lemma 5.4(2),  $\log q \leq L$  and  $\log z \leq L$ , the bracket is  $\ll_A R := x/L^{A+2} + L^2$  in absolute value. Hence, using Lemma 3.26(1),

$$E(x; q) \ll_A \max_{\sqrt{x} \leq z \leq x} |\Delta_\Lambda(z; q, a)| + \frac{R}{\varphi(q)} + \sqrt{x}.$$

*Step 2: summation.* Sum over  $q \leq Q$ : the first terms are bounded by Proposition 10.1, i.e. by  $\ll_A x/L^A$ ; by Lemma 5.3 the second terms sum to  $\ll RL$ ; and the third terms sum to  $\ll Q\sqrt{x} \leq x/L^{A+3} \leq x/L^A$ . Finally

$$RL = \frac{x}{L^{A+1}} + L^3 \leq \frac{x}{L^A} + (A+3)! \frac{x}{L^A},$$

where the last step uses  $L^{A+3}/(A+3)! \leq e^L = x$ .  $\square$

**Theorem 3** (Bombieri–Vinogradov). *Assume the Siegel–Walfisz theorem (Theorem 1) and the large sieve inequality (Theorem 2). Then for every  $A \in \mathbb{N}$ , all real  $x \geq 2$  and all real  $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ ,*

$$\sum_{q \leq Q} \sup_{1 \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A}.$$

*The bound holds whether the suprema are taken in  $[0, \infty]$ , i.e. for  $\sum_{q \leq Q} E(x; q)$ , or in  $\mathbb{R}$ .*

*Proof.* *Step 1:*  $x \leq e^{32}$ . Apply Lemma 10.3.

*Step 2:*  $x > e^{32}$ . Then  $L := \log x \geq 32$ . Put  $U = V := e^{\sqrt{L}}$ . Then  $(x, U, V)$  is a parameter datum (Definition 3.23):  $x \geq 2$ ,  $U, V \geq e^{\sqrt{L}}$  trivially, and  $UV = e^{2\sqrt{L}} \leq e^{L/2} = \sqrt{x}$  because  $2\sqrt{L} \leq L/2$  for  $L \geq 16$ . Now Lemma 10.4 gives the bound for  $\sum_{q \leq Q} E(x; q)$ .

*Step 3: the real form.* Each  $E(x; q)$  with  $q \geq 1$  is finite and equals the corresponding supremum taken in  $\mathbb{R}$  (Definition 10.2), so the bound for  $\sum_{q \leq Q} E(x; q)$  is the same statement as the bound with real suprema.  $\square$

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